Topics in Analysis

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'Topics in Analysis' is a course with no real destination. It is a tour of some of the prettier shells lying on the beach of real and complex analysis, picked up because they are pretty rather than because they are needed to build something else. We will prove that you cannot stir a cup of coffee without leaving some point of the surface where it started, that every continuous function on an interval is a uniform limit of polynomials, that there are continuous functions which are differentiable nowhere at all, and that the number $\sum 10^{-n!}$ satisfies no polynomial equation with integer coefficients. Along the way we will meet a theorem which won its author a Nobel prize in economics.

What ties these together is a small number of extremely powerful ideas – compactness, completeness, and the systematic exploitation of approximation. Almost every argument in the course is a variation on one of the following themes: a continuous function on a compact set attains its bounds; a sequence which does not move very much converges; and if you want to prove that an object with strange properties exists, it is often easier to prove that *almost every* object has them than to write one down.

This article constitutes my notes for the 'Topics in Analysis' course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

Contents

§1 Preliminaries

This first section is mostly a matter of recalling things that we already know, and fixing notation. The course proper begins in the next section; the reader who finds the material below either worryingly familiar or worryingly unfamiliar is asked to suspend judgement until then.

§1.1 Metric Spaces and Completeness

Almost all of the spaces we deal with in this course will be $\mathbb{R}^m$, $\mathbb{C}$, or a space of functions or of compact sets. The common framework is that of a metric space.

Definition 1.1 (Metric Space)

Let X be a non-empty set. A function $d : X^2 \to \mathbb{R}$ is a **metric** on X if

(i) $d(x, y) \geq 0$ for all $x, y \in X$;

(ii) $d(x, y) = 0$ if and only if $x = y$;

(iii) $d(x, y) = d(y, x)$ for all $x, y \in X$;

(iv) $d(x, y) + d(y, z) \geq d(x, z)$ for all $x, y, z \in X$.

We then call (X, d) a **metric space**.

The examples to keep in mind are Euclidean space $\mathbb{R}^m$ with $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$, the complex plane with $d(z, w) = |z - w|$, and the space $C([a, b])$ of continuous real-valued functions on $[a, b]$ with the uniform metric $d(f, g) = \|f - g\|_\infty$. We write

$$B(x, r) = \{y \in X : d(x, y) < r\}, \qquad \overline{B}(x, r) = \{y \in X : d(x, y) \leq r\}$$

for the open and closed balls of radius r about x, and we say $E \subseteq X$ is **closed** if $x_n \in E$ and $x_n \to x$ imply $x \in E$, and $U \subseteq X$ is **open** if for each $u \in U$ there is a $\delta > 0$ with $B(u, \delta) \subseteq U$. These are complementary notions: E is closed if and only if $X \setminus E$ is open.

The single most important property a metric space can have, for our purposes, is completeness.

Definition 1.2 (Cauchy Sequence, Completeness)

Let (X, d) be a metric space. A sequence (a_n) in X is a **Cauchy sequence** if, given $\varepsilon > 0$, there is an N with $d(a_n, a_m) < \varepsilon$ for all $n, m \geq N$. We say (X, d) is **complete** if every Cauchy sequence in X converges.

We will need two small observations about Cauchy sequences over and over again, so it is worth isolating them.

Lemma 1.3

Let (X, d) be a metric space.

(i) If a Cauchy sequence (x_n) has a subsequence converging to x, then $x_n \to x$.

(ii) Suppose $\varepsilon_n > 0$ with $\sum_{n \geq 1} \varepsilon_n < \infty$. If every sequence (x_n) in X with $d(x_n, x_{n+1}) < \varepsilon_n$ for all n converges, then (X, d) is complete.

Proof. (i) Let $\varepsilon > 0$ and pick N with $d(x_n, x_m) < \varepsilon/2$ for $n, m \geq N$. If $x_{n(k)} \to x$, choose k with $n(k) \geq N$ and $d(x_{n(k)}, x) < \varepsilon/2$. Then for $n \geq N$ we have $d(x_n, x) \leq d(x_n, x_{n(k)}) + d(x_{n(k)}, x) < \varepsilon$.

(ii) Let (x_n) be Cauchy. Choose $n(1) < n(2) < \cdots$ so that $d(x_p, x_q) < \varepsilon_k$ whenever $p, q \geq n(k)$. Then the subsequence $y_k = x_{n(k)}$ satisfies $d(y_k, y_{k+1}) < \varepsilon_k$, so converges by hypothesis, and (i) finishes the job. □

In practice we use (ii) with $\varepsilon_n = 2^{-n}$ or $\varepsilon_n = 100^{-n}$: to check completeness it suffices to deal with sequences whose steps shrink geometrically, which is a considerable convenience. We record without proof the fundamental example.

Theorem 1.4

Euclidean space $\mathbb{R}^m$ with the usual metric is complete, as is $\mathbb{C}$.

The proof is the usual one: a Cauchy sequence in $\mathbb{R}^m$ is coordinatewise Cauchy, and $\mathbb{R}$ is complete by the general principle of convergence from Analysis I.

§1.2 The Contraction Mapping Theorem

Completeness earns its keep almost immediately. The following theorem was met in IB Analysis and Topology, but it is worth restating here, both because we shall want it and because it is the model against which everything in Section 3 should be compared.

Definition 1.5 (Contraction)

Let (X, d) be a metric space. A map $f : X \to X$ is a **contraction mapping** if there is a constant $0 \le K < 1$ with

$$d\big(f(x), f(y)\big) \le K\, d(x, y)$$

for all $x, y \in X$.

A contraction is certainly continuous – indeed uniformly so. The point of the definition is that it does not merely fail to increase distances, it shrinks them by a definite factor, and it is that definite factor which makes everything work.

Theorem 1.6 (Contraction Mapping Theorem)

Let (X, d) be a non-empty complete metric space and let $f : X \to X$ be a contraction mapping with constant K. Then f has exactly one fixed point w. Moreover, for any $x_0 \in X$ the iterates $x_{n+1} = f(x_n)$ satisfy $x_n \to w$, with

$$d(x_n, w) \le \frac{K^n}{1-K} d(x_0, x_1).$$

Proof. Uniqueness first, since it is free: if $f(v) = v$ and $f(w) = w$ then $d(v, w) = d(f(v), f(w)) \le K d(v, w)$, and as $K < 1$ this forces $d(v, w) = 0$.

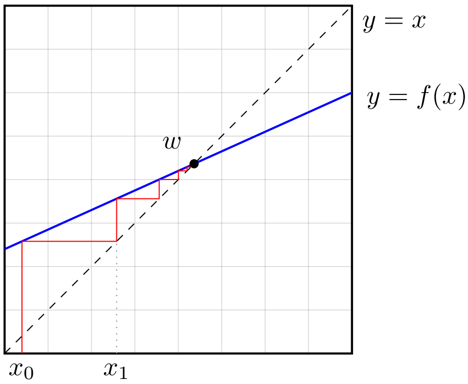
For existence, fix $x_0 \in X$ and set $x_{n+1} = f(x_n)$. An immediate induction gives $d(x_n, x_{n+1}) \le K^n d(x_0, x_1)$, so for $m > n$

$$d(x_n, x_m) \le \sum_{r=n}^{m-1} d(x_r, x_{r+1}) \le d(x_0, x_1) \sum_{r=n}^{\infty} K^r = \frac{K^n}{1-K} d(x_0, x_1).$$

Since $K^n \to 0$, the sequence (x_n) is Cauchy, and by completeness $x_n \to w$ for some

$w \in X$. As f is continuous, $f(w) = \lim f(x_n) = \lim x_{n+1} = w$. Letting $m \to \infty$ in the displayed inequality gives the error bound. $\square$

The proof is worth a second look, because it is the pattern for a great many existence proofs: we do not find w, we manufacture a sequence which is forced to converge and then let the limit be w. The picture is the familiar staircase, or *cobweb*, of the numerical analyst.



Both hypotheses are needed. Completeness cannot be dropped: on $X = (0, 1]$ with the usual metric the map $f(x) = x/2$ is a contraction with $K = 1/2$, but it has no fixed point in X. Nor can $K < 1$ be weakened to $d(f(x), f(y)) < d(x, y)$: the map $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = \sqrt{1 + x^2}$ satisfies

$$|f(x) - f(y)| = \frac{|x^2 - y^2|}{\sqrt{1 + x^2} + \sqrt{1 + y^2}} = |x - y| \cdot \frac{|x + y|}{\sqrt{1 + x^2} + \sqrt{1 + y^2}} < |x - y|$$

for $x \neq y$, yet $f(x) > x$ everywhere, so there is no fixed point at all.

The contraction mapping theorem is enormously useful – it is what makes Picard's existence theorem for differential equations and the inverse function theorem work – but the hypothesis is very strong indeed, and it is easy to write down continuous self-maps of a disc which contract nothing. That such maps still have fixed points is the subject of Section 3, and the price we pay is that the proof will tell us nothing whatever about how to find one.

§1.3 Compactness in Euclidean Space

In this course we deal only with subsets of $\mathbb{R}^m$, and there 'compact' may be taken to mean 'closed and bounded'. (The reader should be warned that this equivalence fails badly in general metric spaces, and that in the world of topological spaces one must further distinguish compactness from sequential compactness. We will not need any of this: if you intend to climb Everest you need your own oxygen supply, but for the Gog Magogs you do not.)

Theorem 1.7 (Bolzano–Weierstrass)

If $K > 0$ and $\mathbf{x}_r \in \mathbb{R}^m$ satisfy $\|\mathbf{x}_r\| \leq K$ for all r, then there exist $\mathbf{x} \in \mathbb{R}^m$ and $r(1) < r(2) < \cdots$ such that $\mathbf{x}_{r(k)} \to \mathbf{x}$.

Proof. For $m = 1$ this is Analysis I. For general m, apply the one-dimensional result to the first coordinates to get a subsequence along which they converge, then pass to a further subsequence along which the second coordinates converge, and so on. After m steps every coordinate converges, and convergence in $\mathbb{R}^m$ is coordinatewise. $\square$

Theorem 1.8

A subset E of $\mathbb{R}^m$ is closed and bounded if and only if every sequence in E has a subsequence converging to a limit in E.

Proof. If E is closed and bounded, Bolzano–Weierstrass gives a convergent subsequence and closedness puts the limit in E. Conversely, if E is unbounded, choose $\mathbf{x}_r \in E$ with $\|\mathbf{x}_r\| \geq r$; no subsequence can converge. If E is not closed, take $\mathbf{x}_r \in E$ with $\mathbf{x}_r \to \mathbf{x} \notin E$; every subsequence converges to $\mathbf{x} \notin E$. $\square$

We shall call closed bounded subsets of $\mathbb{R}^m$ **compact**, and refer to the property in Theorem 1.8 as the Bolzano–Weierstrass property. Its usefulness comes from the following two results, which will be invoked so frequently that we shall often not bother to mention them.

Theorem 1.9

If $E \subseteq \mathbb{R}^m$ is compact and $f : E \to \mathbb{R}^n$ is continuous, then $f(E)$ is compact.

Proof. Let $\mathbf{y}_r = f(\mathbf{x}_r)$ with $\mathbf{x}_r \in E$. Choose a subsequence $\mathbf{x}_{r(k)} \to \mathbf{x} \in E$. By continuity $\mathbf{y}_{r(k)} \to f(\mathbf{x}) \in f(E)$. $\square$

Theorem 1.10

If $E \subseteq \mathbb{R}^m$ is non-empty and compact and $f : E \to \mathbb{R}$ is continuous, then there exist $\mathbf{a}, \mathbf{b} \in E$ with $f(\mathbf{b}) \leq f(\mathbf{x}) \leq f(\mathbf{a})$ for all $\mathbf{x} \in E$.

Proof. $f(E)$ is a non-empty compact subset of $\mathbb{R}$, hence bounded, and contains its supremum and infimum since it is closed. $\square$

We will also want uniform continuity, which is compactness in another guise.

Definition 1.11 (Uniform Continuity)

Let (X, d) and (Y, ρ) be metric spaces. A function $f : X \to Y$ is **uniformly continuous** if, given $\varepsilon > 0$, we can find $\delta > 0$ such that $\rho(f(a), f(b)) < \varepsilon$ whenever $d(a, b) < \delta$.

Theorem 1.12

If E is a compact subset of $\mathbb{R}^m$ and $f : E \to \mathbb{R}^p$ is continuous, then f is uniformly continuous.

Proof. If not, there is an $\varepsilon > 0$ and points $\mathbf{a}_n, \mathbf{b}_n \in E$ with $\|\mathbf{a}_n - \mathbf{b}_n\| < 1/n$ but $\|f(\mathbf{a}_n) - f(\mathbf{b}_n)\| \geq \varepsilon$. Pass to a subsequence with $\mathbf{a}_{n(k)} \to \mathbf{a} \in E$; then $\mathbf{b}_{n(k)} \to \mathbf{a}$ too, so both $f(\mathbf{a}_{n(k)})$ and $f(\mathbf{b}_{n(k)})$ tend to $f(\mathbf{a})$, contradicting $\left\|f(\mathbf{a}_{n(k)}) - f(\mathbf{b}_{n(k)})\right\| \geq \varepsilon$. □

Finally, a piece of notation which will be with us all course. If A is a non-empty subset of a metric space and x a point, we write

$$d(x, A) = \inf\{d(x, a) : a \in A\}.$$

Lemma 1.13

Let A be a non-empty subset of a metric space (X, d). Then $x \mapsto d(x, A)$ satisfies $|d(x, A) - d(y, A)| \leq d(x, y)$, and so is (uniformly) continuous. If A is closed, then $d(x, A) = 0$ if and only if $x \in A$.

Proof. For any $a \in A$ we have $d(x, A) \leq d(x, a) \leq d(x, y) + d(y, a)$; taking the infimum over a gives $d(x, A) \leq d(x, y) + d(y, A)$, and symmetry gives the Lipschitz bound. If $d(x, A) = 0$ we can find $a_n \in A$ with $d(x, a_n) \to 0$, so $a_n \to x$ and $x \in A$ when A is closed. □

§1.4 The Fundamental Theorem of Algebra

Theorem 1.10 looks like a piece of abstract nonsense, but it is enough to prove a theorem which – contrary to its name – is a theorem of analysis.

Theorem 1.14 (Fundamental Theorem of Algebra)

Every non-constant polynomial P with complex coefficients has a root in $\mathbb{C}$.

Proof. Write $P(z) = \sum_{j=0}^{n} a_j z^j$ with $n \geq 1$, and (dividing through) assume $a_n = 1$. Then for $z \neq 0$,

$$|P(z)| \geq |z|^n \left(1 - |a_{n-1}||z|^{-1} - \cdots - |a_0||z|^{-n}\right),$$

so if we set $R = 2\left(2 + \sum_{j=0}^{n-1} |a_j|\right)$ then $|z| \geq R$ gives $|P(z)| \geq |z|^n/2 \geq R^n/2 > |a_0| = |P(0)|$.

Now $\overline{B}(0, R)$ is compact and $|P|$ is continuous, so by Theorem 1.10 there is a $z_0 \in \overline{B}(0, R)$ at which $|P|$ attains its minimum over $\overline{B}(0, R)$. By the previous paragraph $|P(z_0)| \leq |P(0)| < |P(z)|$ for all $|z| \geq R$, so in fact z_0 is a *global* minimum of $|P|$ on $\mathbb{C}$.

Suppose, for contradiction, that $P(z_0) \neq 0$. Replacing $P(z)$ by $P(z + z_0)$ we may assume $z_0 = 0$, and multiplying P by a suitable unimodular constant we may assume $a_0 = P(0) > 0$. Let $m \geq 1$ be least with $a_m \neq 0$, so that

$$P(z) = a_0 + a_m z^m + \sum_{j=m+1}^{n} a_j z^j.$$

Choosing ω with $\omega^m = -|a_m|/a_m$ and replacing $P(z)$ by $P(\omega z)$, we may further assume $a_m < 0$. Now for real $\delta > 0$,

$$\frac{P(\delta) - P(0)}{\delta^m} = a_m + \sum_{j=m+1}^{n} a_j \delta^{j-m} \to a_m < 0$$

as $\delta \to 0^+$. So for small positive δ the real part of $P(\delta) - P(0)$ is negative and its imaginary part is $o(\delta^m)$, whence $|P(\delta)| < |P(0)|$. This contradicts the minimality of $|P(0)|$, so $P(z_0) = 0$. □

We shall meet a completely different proof of Theorem 1.14, using winding numbers, in the final section of the course. In the meantime we record the standard consequences, which we use freely.

Lemma 1.15

Let P be a polynomial of degree n over $\mathbb{C}$.

(i) If $P(a) = 0$ then $P(z) = (z - a)Q(z)$ for some polynomial Q of degree $n - 1$.

(ii) $P(z) = A\prod_{j=1}^{n}(z - a_j)$ for some $A \neq 0$ and $a_j \in \mathbb{C}$.

(iii) If P has degree at most n and vanishes at $n + 1$ distinct points, then $P = 0$.

Proof. (i) By induction on n, division gives $P(z) = (z-a)Q(z)+r$ with $\deg Q = n-1$ and $r \in \mathbb{C}$; putting $z = a$ gives $r = 0$. (ii) follows by induction using Theorem 1.14 and (i). (iii) If $P \neq 0$ has degree $d \leq n$ then by (ii) it has at most $d \leq n$ roots. □

§2 Laplace's Equation

Our first genuine topic is a piece of analysis which sits behind a great deal of the theory of partial differential equations. Recall that if ϕ is a real-valued function on (an open subset of) $\mathbb{R}^m$ with enough derivatives, we write

$$\nabla^2\phi = \sum_{j=1}^{m} \frac{\partial^2\phi}{\partial x_j^2},$$

and **Laplace's equation** is the equation $\nabla^2\phi = 0$. Solutions are called *harmonic*. In Part IA you cheerfully solved boundary value problems for Laplace's equation on discs and rectangles, and you may have been left with the impression that the Dirichlet problem

$$\nabla^2\phi = 0 \quad \text{on } \Omega, \qquad \phi = f \quad \text{on } \partial\Omega$$

always has exactly one solution. We shall see that uniqueness holds in great generality, but that existence can fail.

We first need to recall three pieces of point-set vocabulary.

Definition 2.1 (Interior, Closure, Boundary)

Let (X, d) be a metric space and $E \subseteq X$.

(i) The **interior** $\operatorname{Int} E$ is the set of $x \in X$ for which there is a $\delta > 0$ with

$B(x, \delta) \subseteq E$.

(ii) The **closure** $\mathrm{Cl}\, E$ is the set of $x \in X$ for which there are $e_n \in E$ with $e_n \to x$.

(iii) The **boundary** of E is $\partial E = \mathrm{Cl}\, E \setminus \mathrm{Int}\, E$.

It is easy to check that $\mathrm{Int}\, E$ is the largest open set contained in E, that $\mathrm{Cl}\, E$ is the smallest closed set containing E, and hence that $\partial E = \mathrm{Cl}\, E \cap \mathrm{Cl}(X \setminus E)$ is closed. If $E \subseteq \mathbb{R}^m$ is bounded then so is $\mathrm{Cl}\, E$, so $\mathrm{Cl}\, E$ and ∂E are compact.

§2.1 The Maximum Principle

Throughout this subsection Ω is a non-empty bounded open subset of $\mathbb{R}^m$ and $\phi : \mathrm{Cl}\, \Omega \to \mathbb{R}$ is continuous and twice differentiable on Ω. Note that $\mathrm{Cl}\, \Omega$ is compact and non-empty, so ϕ does attain a maximum somewhere; the content of what follows is *where*.

The following lemma is really just the second derivative test, applied one coordinate at a time.

Lemma 2.2

If $\nabla^2 \phi > 0$ everywhere on Ω, then ϕ does not attain its maximum at any point of Ω.

Proof. Suppose ϕ attains its maximum over $\mathrm{Cl}\, \Omega$ at $\mathbf{x}^* \in \Omega$. Since Ω is open there is a $\delta > 0$ with

$$(x_1^* - \delta, x_1^* + \delta) \times \cdots \times (x_m^* - \delta, x_m^* + \delta) \subseteq \Omega.$$

For each j define $g_j(t) = \phi(x_1^*, \dots, x_{j-1}^*, x_j^* + t, x_{j+1}^*, \dots, x_m^*)$ for $|t| < \delta$. Then g_j is twice differentiable with a maximum at $t = 0$, so $g_j'(0) = 0$ and $g_j''(0) \le 0$. But $g_j''(0) = \partial^2 \phi / \partial x_j^2$ evaluated at $\mathbf{x}^*$, so summing over j gives $\nabla^2 \phi(\mathbf{x}^*) \le 0$, a contradiction. $\square$

The hypothesis $\nabla^2 \phi > 0$ is of course much stronger than what we want. The trick, which is the same one used to deduce Rolle's theorem from the fact that an interior maximum is stationary, is to add a small strictly subharmonic perturbation and then let it go to zero.

Theorem 2.3 (Maximum Principle)

If $\nabla^2 \phi = 0$ on Ω, then ϕ attains its maximum over $\mathrm{Cl}\, \Omega$ at some point of $\partial \Omega$.

Proof. For $n \ge 1$ set $\phi_n(\mathbf{x}) = \phi(\mathbf{x}) + \frac{1}{n} \sum_{j=1}^m x_j^2$. Then $\nabla^2 \phi_n = 2m/n > 0$ on Ω, so by Lemma 2.2 the maximum of ϕ_n over $\mathrm{Cl}\, \Omega$ is attained at some $\mathbf{x}_n^* \in \partial \Omega$. Since $\partial \Omega$ is compact, we may pass to a subsequence $\mathbf{x}_{n(j)}^* \to \mathbf{x}^* \in \partial \Omega$.

Let R be such that $\|\mathbf{x}\| \le R$ for all $\mathbf{x} \in \mathrm{Cl}\, \Omega$. For every $\mathbf{x} \in \mathrm{Cl}\, \Omega$ we have

$$\phi(\mathbf{x}_{n(j)}^*) + \frac{R^2}{n(j)} \ge \phi_{n(j)}(\mathbf{x}_{n(j)}^*) \ge \phi_{n(j)}(\mathbf{x}) \ge \phi(\mathbf{x}).$$

Letting $j \to \infty$ and using continuity of ϕ gives $\phi(\mathbf{x}^*) \ge \phi(\mathbf{x})$, as required. $\square$

Applying the theorem to $-\phi$ shows that the minimum is also attained on the boundary, and uniqueness follows at once.

Theorem 2.4 (Uniqueness for the Dirichlet Problem)

Suppose $\phi, \psi : \operatorname{Cl}\Omega \to \mathbb{R}$ are continuous, twice differentiable on Ω, and satisfy $\nabla^2\phi = \nabla^2\psi = 0$ on Ω. If $\phi = \psi$ on $\partial\Omega$, then $\phi = \psi$ on $\operatorname{Cl}\Omega$.

Proof. Set $\chi = \phi - \psi$. Then $\nabla^2\chi = 0$ on Ω and $\chi = 0$ on $\partial\Omega$, so by Theorem 2.3 the maximum of χ is 0, and applying the same to $-\chi$ gives that the minimum is 0. Hence $\chi \equiv 0$. □

Remark. The same argument gives a maximum principle for analytic functions. If $\Omega \subseteq \mathbb{C}$ is bounded and open and $f : \operatorname{Cl}\Omega \to \mathbb{C}$ is continuous and analytic on Ω, then $\operatorname{Re} f$ and $\operatorname{Im} f$ are harmonic, and one deduces that $|f|$ attains its maximum on $\partial\Omega$: given z_0, rotate f so that $\operatorname{Re} f(z_0) = |f(z_0)|$ and apply Theorem 2.3 to $\operatorname{Re} f$.

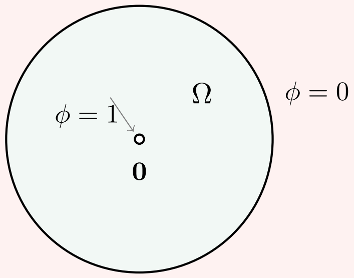
§2.2 Zaremba's Example

So much for uniqueness. What about existence? The following example, due to Zaremba, shows that we cannot expect a solution for every bounded open Ω and every continuous boundary datum. It is a nice illustration of how uniqueness theorems can be used to *destroy* existence.

Example 2.5 (Zaremba)

Let $D = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| < 1\}$ and let $\Omega = D \setminus \{\mathbf{0}\}$ be the punctured disc. Then Ω is open and bounded, $\operatorname{Cl}\Omega = \operatorname{Cl} D$, and

$$\partial\Omega = \{\mathbf{x} : \|\mathbf{x}\| = 1\} \cup \{\mathbf{0}\}.$$



We claim that there is no continuous $\phi : \operatorname{Cl}\Omega \to \mathbb{R}$, twice differentiable on Ω, with $\nabla^2\phi = 0$ on Ω and

$$\phi(\mathbf{x}) = \begin{cases} 0 & \|\mathbf{x}\| = 1, \\ 1 & \mathbf{x} = \mathbf{0}. \end{cases}$$

Suppose such a ϕ existed. Let R_θ be the rotation of $\mathbb{R}^2$ through θ about the origin. Then $\mathbf{x} \mapsto \phi(R_\theta \mathbf{x})$ is also continuous on $\operatorname{Cl}\Omega$, harmonic on Ω, and takes the *same* boundary values (since the boundary data are rotation-invariant). By the uniqueness Theorem 2.4 we get $\phi = \phi \circ R_\theta$ for every θ, so ϕ is radially symmetric: $\phi(\mathbf{x}) = f(\|\mathbf{x}\|)$ for some $f : [0, 1] \to \mathbb{R}$.

Writing $r = \sqrt{x^2 + y^2}$ and computing,

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= \frac{x}{r} f'(r), \\ \frac{\partial^2 \phi}{\partial x^2} &= \left(\frac{1}{r} - \frac{x^2}{r^3} \right) f'(r) + \frac{x^2}{r^2} f''(r), \end{aligned}$$

and similarly for y, so that adding gives the familiar radial form

$$\nabla^2 \phi = f''(r) + \frac{1}{r} f'(r) = \frac{1}{r} \frac{\mathrm{d}}{\mathrm{d}r} \big(r f'(r) \big).$$

Thus $\frac{\mathrm{d}}{\mathrm{d}r}(rf'(r)) = 0$ for $0 < r < 1$, so $rf'(r) = B$ and $f(r) = A + B \log r$ for constants A, B. But ϕ is continuous at $\mathbf{0}$, and $A + B \log r$ has a limit as $r \to 0^+$ only when $B = 0$. So f is constant, and ϕ cannot take the value 0 on $\|\mathbf{x}\| = 1$ and 1 at $\mathbf{0}$. This is the contradiction we wanted.

The moral is that the boundary point $\mathbf{0}$ is 'too thin' for the boundary condition there to be felt. Lebesgue later produced a three-dimensional example (the *Lebesgue thorn*) in which the offending boundary point is a cusp rather than an isolated point, so that the failure is not simply a matter of the boundary having an isolated point. Deciding exactly which boundary points are 'regular' in this sense is a substantial piece of potential theory, and we shall not pursue it.

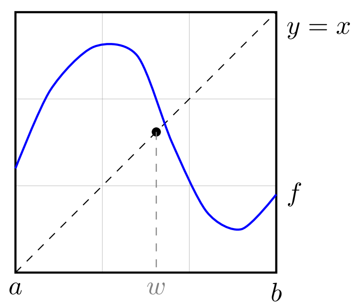
§3 Fixed Point Theorems

In Analysis I we proved the intermediate value theorem, and used it to prove a very pretty one-dimensional fixed point theorem.

Theorem 3.1
If $f : [a, b] \to [a, b]$ is continuous, then there is a $w \in [a, b]$ with $f(w) = w$.

Proof. Set $g(x) = f(x) - x$. Then g is continuous, $g(a) = f(a) - a \geq 0$ and $g(b) = f(b) - b \leq 0$. By the intermediate value theorem there is a $w \in [a, b]$ with $g(w) = 0$. $\square$

The picture is irresistible: the graph of f lives in the square $[a, b]^2$, it starts on or above the diagonal and ends on or below it, so it must cross.



The implication can be reversed: Theorem 3.1 implies the intermediate value theorem, and both are equivalent to the statement that $[0, 1]$ is connected. This is worth bearing in mind, because the object of this section is to prove the two-dimensional analogue, and the proof will consist almost entirely of showing that a chain of superficially unrelated statements are all equivalent to one another.

Theorem 3.2 (Brouwer's Fixed Point Theorem)
Let $\overline{D} = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 1\}$. If $f : \overline{D} \to \overline{D}$ is continuous, then there is a $\mathbf{w} \in \overline{D}$ with $f(\mathbf{w}) = \mathbf{w}$.

We shall keep strictly to two dimensions, but the reader should know that the theorem and most of its consequences hold in $\mathbb{R}^n$; the only ingredient which needs real work to generalise is Sperner's lemma, and even there the changes are cosmetic rather than conceptual.

The result is not restricted to discs. The following observation lets us move it around freely, and we shall use it without comment.

Lemma 3.3
Suppose $g : \overline{D} \to A$ is a bijection with g and g^{-1} continuous. If $F : A \to A$ is continuous, then F has a fixed point.

Proof. The map $g^{-1} \circ F \circ g : \overline{D} \to \overline{D}$ is continuous, so by Theorem 3.2 has a fixed point $\mathbf{w}$. Then $g(\mathbf{w})$ is a fixed point of F. $\square$

In particular Brouwer's theorem holds for closed squares, closed triangles, and any other set homeomorphic to a disc. Throughout we write

$$D = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| < 1\}, \quad \overline{D} = \{\mathbf{x} : \|\mathbf{x}\| \leq 1\}, \quad \partial D = \{\mathbf{x} : \|\mathbf{x}\| = 1\}.$$

§3.1 The No Retraction Theorem

The first equivalence is the one which explains why the theorem is true: a fixed-point-free map lets you comb the disc onto its boundary, and that is something you cannot do.

Theorem 3.4
The following are equivalent.

(i) Every continuous $f : \overline{D} \to \overline{D}$ has a fixed point.

(ii) There is no continuous $g : \overline{D} \to \partial D$ with $g(\mathbf{x}) = \mathbf{x}$ for all $\mathbf{x} \in \partial D$. (Such a g is called a **retraction**.)

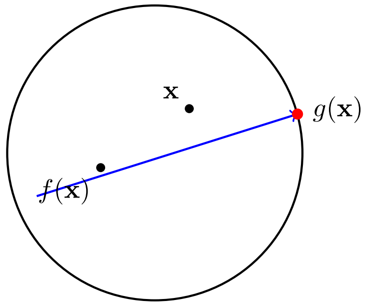
Proof. Suppose (i) holds and $g : \overline{D} \to \partial D$ is a retraction. Let R be rotation through π about $\mathbf{0}$. Then $R \circ g : \overline{D} \to \overline{D}$ is continuous, so has a fixed point $\mathbf{w}$. But $R(g(\mathbf{w})) = \mathbf{w}$ forces $\mathbf{w} \in R(\partial D) = \partial D$, and then $g(\mathbf{w}) = \mathbf{w}$, so $R\mathbf{w} = \mathbf{w}$, which is false for $\mathbf{w} \neq \mathbf{0}$. Since $\mathbf{0} \notin \partial D$ we have a contradiction, and (ii) holds.

Conversely, suppose (ii) holds and $f : \overline{D} \to \overline{D}$ is continuous with no fixed point. For $\mathbf{x} \neq \mathbf{y}$ in $\overline{D}$ let $F(\mathbf{x}, \mathbf{y})$ be the point where the ray starting at $\mathbf{y}$ and passing through

$\mathbf{x}$ meets ∂D; explicitly, writing $\mathbf{u} = (\mathbf{x} - \mathbf{y})/\|\mathbf{x} - \mathbf{y}\|$, we have $F(\mathbf{x}, \mathbf{y}) = \mathbf{x} + t\mathbf{u}$ where $t \geq 0$ is the unique non-negative root of $\|\mathbf{x} + t\mathbf{u}\|^2 = 1$, that is

$$t = -\mathbf{x} \cdot \mathbf{u} + \sqrt{(\mathbf{x} \cdot \mathbf{u})^2 + 1 - \|\mathbf{x}\|^2}.$$

This is a continuous function of $(\mathbf{x}, \mathbf{y})$ on $\{(\mathbf{x}, \mathbf{y}) \in \overline{D}^2 : \mathbf{x} \neq \mathbf{y}\}$. Set $g(\mathbf{x}) = F(\mathbf{x}, f(\mathbf{x}))$, which makes sense since $f(\mathbf{x}) \neq \mathbf{x}$ everywhere. Then g is continuous, $g(\overline{D}) \subseteq \partial D$, and if $\mathbf{x} \in \partial D$ then $\mathbf{x}$ itself is already on the ray and on ∂D, so $g(\mathbf{x}) = \mathbf{x}$. Thus g is a retraction, contradicting (ii). $\square$

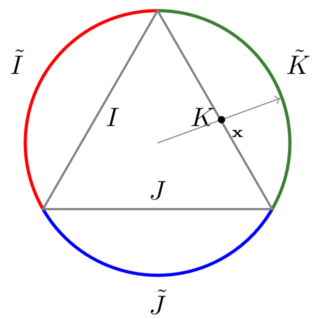


§3.2 From Retractions to Colourings

The next step replaces the circle by a triangle and the retraction by a map which preserves three closed pieces of the boundary. Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ be unit vectors making angles of $2\pi/3$ with each other, let T be the closed triangle with these vertices, and let I, J, K be its three closed sides. Correspondingly let

$$\tilde{I} = \{(\cos\theta, \sin\theta) : 0 \leq \theta \leq \tfrac{2\pi}{3}\}, \quad \tilde{J} = \{(\cos\theta, \sin\theta) : \tfrac{2\pi}{3} \leq \theta \leq \tfrac{4\pi}{3}\},$$

and $\tilde{K} = \{(\cos\theta, \sin\theta) : \frac{4\pi}{3} \leq \theta \leq 2\pi\}$ be the three corresponding closed arcs of ∂D.



Lemma 3.5

The following three statements are equivalent.

(i) There is no retraction from $\overline{D}$ to ∂D.

(ii) There is no continuous $\tilde{k} : \overline{D} \to \partial D$ with $\tilde{k}(\tilde{I}) \subseteq \tilde{I}$, $\tilde{k}(\tilde{J}) \subseteq \tilde{J}$ and $\tilde{k}(\tilde{K}) \subseteq \tilde{K}$.

(iii) There is no continuous $k : T \to \partial T$ with $k(I) \subseteq I$, $k(J) \subseteq J$ and $k(K) \subseteq K$.

Proof. (ii) $\Rightarrow$ (i) is immediate, since a retraction certainly maps each arc into itself. For (i) $\Rightarrow$ (ii), suppose $\tilde{k}$ is as in (ii) and let R be rotation through π. Then $R \circ \tilde{k} : \overline{D} \to \overline{D}$ has no fixed point: a fixed point $\mathbf{w}$ would have to lie on ∂D, and $\mathbf{w}$ lies in one of the three arcs, say $\tilde{I}$, so $\tilde{k}(\mathbf{w}) \in \tilde{I}$ while $R\tilde{k}(\mathbf{w}) = \mathbf{w} \in \tilde{I}$ means $\tilde{k}(\mathbf{w}) = R\mathbf{w}$, which lies in the arc antipodal to the one containing $\mathbf{w}$. Since an arc of length $2\pi/3$ is disjoint from its antipodal image, this is impossible. By Theorem 3.4 this contradicts (i).

For the equivalence of (ii) and (iii), define $\Phi : \overline{D} \to T$ by $\Phi(\mathbf{0}) = \mathbf{0}$ and, for $\mathbf{x} \neq \mathbf{0}$, $\Phi(\mathbf{x}) = \rho(\mathbf{x})\mathbf{x}$ where $\rho(\mathbf{x}) > 0$ is chosen so that $\rho(\mathbf{x})\mathbf{x} \in \partial T$ when $\|\mathbf{x}\| = 1$; in other words we stretch each radius of the disc onto the corresponding radius of the triangle. Since $\mathbf{0}$ is interior to T and ∂T meets each ray from $\mathbf{0}$ exactly once, ρ is a continuous positive function of the direction of $\mathbf{x}$ alone, and Φ is a homeomorphism carrying ∂D onto ∂T and $\tilde{I}, \tilde{J}, \tilde{K}$ onto I, J, K respectively. So $\tilde{k} \mapsto \Phi \circ \tilde{k} \circ \Phi^{-1}$ is a bijection between maps of the type in (ii) and maps of the type in (iii). □

The next equivalence is the crucial one: it converts a statement about continuous maps into a statement about closed sets, and closed sets are things we can attack combinatorially.

Lemma 3.6

The following are equivalent.

(i) There is no continuous $h : T \to \partial T$ with $h(I) \subseteq I$, $h(J) \subseteq J$, $h(K) \subseteq K$.

(ii) If A, B, C are closed subsets of T with $A \supseteq I$, $B \supseteq J$, $C \supseteq K$ and $A \cup B \cup C = T$, then $A \cap B \cap C \neq \emptyset$.

Proof. Suppose (ii) holds and h is as in (i). Put $A = h^{-1}(I)$, $B = h^{-1}(J)$, $C = h^{-1}(K)$. These are closed (preimages of closed sets under a continuous map), they cover T since $I \cup J \cup K = \partial T$, and $A \supseteq I$ since $h(I) \subseteq I$; similarly for B and C. But $I \cap J \cap K = \emptyset$ (the three sides have no common point), so $A \cap B \cap C = h^{-1}(I \cap J \cap K) = \emptyset$, contradicting (ii).

Conversely, suppose (i) holds and A, B, C are closed sets as in (ii) with $A \cap B \cap C = \emptyset$. It is convenient to realise T in barycentric coordinates,[a] identifying it with

$$T = \{(x, y, z) \in \mathbb{R}^3 : x, y, z \geq 0,\ x + y + z = 1\},$$

with $I = \{x = 0\} \cap T$, $J = \{y = 0\} \cap T$ and $K = \{z = 0\} \cap T$. Define

$$h(\mathbf{x}) = \frac{1}{d(\mathbf{x}, A) + d(\mathbf{x}, B) + d(\mathbf{x}, C)} \big(d(\mathbf{x}, A),\ d(\mathbf{x}, B),\ d(\mathbf{x}, C)\big).$$

The denominator never vanishes: if it did, then $\mathbf{x}$ would be at zero distance from each of the closed sets A, B, C, so $\mathbf{x} \in A \cap B \cap C = \emptyset$. By Lemma 1.13 each numerator is continuous, so h is continuous, and clearly $h(\mathbf{x}) \in T$ with at least one coordinate zero (as $A \cup B \cup C = T$ means every $\mathbf{x}$ lies in one of them), so $h(T) \subseteq \partial T$. Finally if $\mathbf{x} \in I$ then $\mathbf{x} \in A$, so $d(\mathbf{x}, A) = 0$ and $h(\mathbf{x}) \in I$; similarly for J and K.

This contradicts (i). □

[a] Barycentric coordinates were invented by Möbius, and are exactly the right language for everything in this section.

Putting Theorem 3.4 and Lemmas 3.5 and 3.6 together,

$$\text{Brouwer} \iff \text{no retraction} \iff \text{three arcs} \iff \text{three sides} \iff \text{three colours},$$

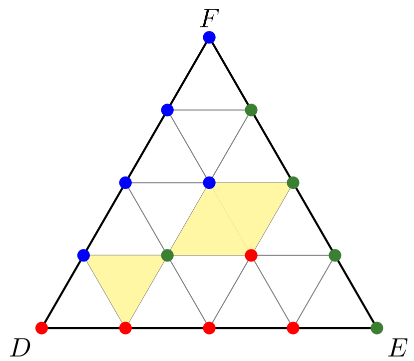
so it is enough to prove the three colours statement. That is what we now do, and the engine is a purely combinatorial lemma of Sperner.

§3.3 Sperner's Lemma

Lemma 3.7 (Sperner)

Consider a triangle DEF divided into a triangular grid by $n-1$ equally spaced lines parallel to each side. Colour every vertex of the grid red, green or blue in such a way that every vertex on the side DE except E is red, every vertex on EF except F is green, and every vertex on FD except D is blue. Then some small triangle of the grid has all three of its vertices differently coloured.

In fact the proof gives more: the number of such 'rainbow' triangles is odd. Here is a grid with $n = 4$; three of the small triangles are rainbow.



Proof. Give each side of each small triangle a value, according to the following rule. Traversing the side in a fixed rotational sense (say clockwise) around its small triangle, set

$$\theta(RG) = \theta(GB) = \theta(BR) = 1, \qquad \theta(GR) = \theta(BG) = \theta(RB) = -1,$$

and $\theta = 0$ on a side whose two endpoints have the same colour. Note that reversing the direction of travel changes the sign of θ.

For a small triangle Δ let $\psi(\Delta)$ be the sum of the values of its three sides, traversed clockwise. If the three vertices of Δ are not all different then two of them agree, and a moment's checking shows $\psi(\Delta) = 0$: for example a triangle coloured R, R, G contributes $0 + 1 + (-1) = 0$. If the vertices are all different then going round clockwise we see either R, G, B or R, B, G, giving $\psi(\Delta) = 3$ or $\psi(\Delta) = -3$. In particular $\psi(\Delta) \neq 0$ exactly when Δ is rainbow, and $\psi(\Delta) \equiv 3 \pmod 6$ in that case.

Now sum $\psi(\Delta)$ over all small triangles Δ. Every interior side of the grid belongs to exactly two small triangles, and is traversed in opposite directions in the two, so its two contributions cancel. Hence

$$\sum_{\Delta} \psi(\Delta) = \sum_{\text{sides on } \partial(DEF)} \theta(\text{side}),$$

the boundary sides being traversed clockwise around the big triangle.

Consider the side DE. All its grid vertices are red except the last, E, which is green. So travelling from D to E we see $R, R, \ldots, R, G$ and the total is $\theta(RG) = 1$. Similarly EF reads $G, G, \ldots, G, B$ contributing 1, and FD reads $B, B, \ldots, B, R$ contributing 1. Hence $\sum_{\Delta} \psi(\Delta) = 3 \neq 0$, so at least one small triangle is rainbow. Indeed, since each rainbow triangle contributes ± 3 and the total is 3, the number of rainbow triangles is odd. □

Proof of Lemma 3.6 (ii), and hence of Brouwer's theorem. Let A, B, C be closed subsets of T with $A \supseteq I$, $B \supseteq J$, $C \supseteq K$ and $A \cup B \cup C = T$, where we label the sides so that $I = DE$, $J = EF$ and $K = FD$.

Fix n and subdivide T into a triangular grid with $n-1$ lines parallel to each side, so that each small triangle has diameter at most $\operatorname{diam}(T)/n$. Colour each grid vertex $\mathbf{v}$ as follows: if $\mathbf{v} \in A$ colour it red, otherwise if $\mathbf{v} \in B$ colour it green, otherwise colour it blue (which is legitimate as $A \cup B \cup C = T$). Since $I \subseteq A$, every vertex on I is red; since $J \subseteq B$, every vertex of J not already coloured red is green – and the only vertex of J lying in I is the shared corner E. Rather than fuss over corners, note that we are free to recolour the three corners: D red, E green, F blue is consistent with $D \in I \subseteq A$, $E \in J \subseteq B$, $F \in K \subseteq C$. Thus the hypotheses of Sperner's lemma hold.

So for each n we obtain a small triangle with vertices $\mathbf{a}_n$ (red, so in A), $\mathbf{b}_n$ (green, so in B) and $\mathbf{c}_n$ (blue, so in C), with

$$\|\mathbf{a}_n - \mathbf{b}_n\|, \ \|\mathbf{b}_n - \mathbf{c}_n\|, \ \|\mathbf{c}_n - \mathbf{a}_n\| \leq \frac{\operatorname{diam}(T)}{n}.$$

As T is compact we may pass to a subsequence along which $\mathbf{a}_{n(j)} \to \mathbf{x}$ for some $\mathbf{x} \in T$. Since the three points of each triple are within $\operatorname{diam}(T)/n$ of each other, $\mathbf{b}_{n(j)} \to \mathbf{x}$ and $\mathbf{c}_{n(j)} \to \mathbf{x}$ too. As A, B, C are closed, $\mathbf{x} \in A \cap B \cap C$, which is therefore non-empty. □

The reader who is worried about the one-dimensional analogue may like to check that the whole of this section degenerates, when T is replaced by $[0, 1]$ and three colours by two, into the following statement: if we colour $0, 1/n, \ldots, 1$ red or blue with 0 red and 1 blue, then some pair of neighbours have different colours. That statement proves the intermediate value theorem, and Sperner's lemma is exactly its two-dimensional cousin.

§3.4 Two Applications

Brouwer's theorem is deep, and it is worth seeing that it does real work. Here is a result which the Markov chains course makes one want very much.

Proposition 3.8 (Perron–Frobenius, stochastic case)

Let $A = (a_{ij})$ be an $n \times n$ matrix with $a_{ij} \geq 0$ for all i, j and $\sum_{i=1}^{n} a_{ij} = 1$ for each j. Then A has an eigenvector with eigenvalue 1 lying in

$$\Sigma = \left\{ \mathbf{x} \in \mathbb{R}^n : x_j \geq 0 \text{ for all } j,\ \textstyle\sum_j x_j = 1 \right\}.$$

Proof. If $\mathbf{x} \in \Sigma$ then $(A\mathbf{x})_i = \sum_j a_{ij} x_j \geq 0$ and

$$\sum_i (A\mathbf{x})_i = \sum_j x_j \sum_i a_{ij} = \sum_j x_j = 1,$$

so $A\Sigma \subseteq \Sigma$. Now Σ is a closed bounded convex subset of the hyperplane $\sum_j x_j = 1$ with non-empty interior in that hyperplane, so is homeomorphic to a closed disc of dimension $n-1$ (radially, as in Lemma 3.5). Applying Brouwer's theorem in dimension $n-1$ via Lemma 3.3, the continuous map $\mathbf{x} \mapsto A\mathbf{x}$ has a fixed point $\mathbf{x}^* \in \Sigma$, and $A\mathbf{x}^* = \mathbf{x}^*$ is the required eigenvector. $\square$

Reading 'i' and 'j' the other way round, this says that every finite state Markov chain has an invariant distribution – a result which is unpleasant to prove by bare hands, and which becomes almost obvious once we are allowed a fixed point theorem.

The second application is a statement which is 'obvious' in the worst sense of the word.

Proposition 3.9

Let $ABCD$ be a square. If γ is a continuous path in the square joining A to C, and τ is a continuous path in the square joining B to D, then γ and τ intersect.

We shall not give the details (they form a rather fiddly exercise), but the idea is worth recording: take the square to be $[-1,1]^2$ with $\gamma, \tau : [0,1] \to [-1,1]^2$, and suppose $\gamma(s) \neq \tau(t)$ for all s, t. Then

$$F(s,t) = \frac{\big(\tau_1(t) - \gamma_1(s),\ \gamma_2(s) - \tau_2(t)\big)}{\|\gamma(s) - \tau(t)\|_\infty}$$

defines a continuous map of $[0,1]^2$ into itself, and one checks that a fixed point of F would have to lie in a corner, where the boundary conditions on γ and τ make it impossible. This is a baby case of the Jordan curve theorem, which asserts that a simple closed curve in the plane divides it into exactly two components. The Jordan curve theorem is genuinely hard, and is not part of this course; but it is comforting to know that its plausibility is not a delusion.

§4 Non-Zero Sum Games

It is said that converting the front garden of a house into a parking space raises the value of that house and lowers the value of every other house on the road. Once everybody has done the conversion, every house is worth less than it was before anybody started. Situations of this kind – where individually rational choices produce a collectively poor outcome – are the meat of game theory, and it turns out that Brouwer's theorem has a great deal to say about them.

Let us set up the simplest possible model. Albert has a choice between actions A_1 and A_2, and Bertha between B_1 and B_2. If Albert plays A_i and Bertha plays B_j, then Albert receives a_{ij} and Bertha receives b_{ij}. In IB Optimisation you studied the *zero-sum* case $a_{ij} = -b_{ij}$, where one player's gain is the other's loss; we shall make no such assumption.

Suppose Albert plays A_i with probability p_i and Bertha plays B_j with probability q_j, where $p_1 + p_2 = q_1 + q_2 = 1$ and all entries are non-negative. Writing $\mathbf{p} = (p_1, p_2)$ and $\mathbf{q} = (q_1, q_2)$, the expected payoffs are

$$A(\mathbf{p}, \mathbf{q}) = \sum_{i=1}^{2}\sum_{j=1}^{2} a_{ij}p_iq_j, \qquad B(\mathbf{p}, \mathbf{q}) = \sum_{i=1}^{2}\sum_{j=1}^{2} b_{ij}p_iq_j.$$

In the zero-sum case you saw that there is a pair of strategies such that neither player would change their choice even knowing the other's. Away from zero-sum we can imagine an endless dance: Albert chooses $\mathbf{p}$, Bertha responds with $\mathbf{q}$, Albert switches to $\mathbf{p}'$, Bertha to $\mathbf{q}'$, and so on. Does the dance ever stop?

Definition 4.1 (Nash Equilibrium)

A pair $(\mathbf{p}^*, \mathbf{q}^*)$ of strategies is a **Nash equilibrium** (or **Nash stable point**) if

$$A(\mathbf{p}^*, \mathbf{q}^*) \geq A(\mathbf{p}, \mathbf{q}^*) \quad \text{and} \quad B(\mathbf{p}^*, \mathbf{q}^*) \geq B(\mathbf{p}^*, \mathbf{q})$$

for all strategies $\mathbf{p}, \mathbf{q}$. In other words, neither player can do better by unilaterally changing their strategy.

Theorem 4.2 (Nash)

For any real numbers a_{ij}, b_{ij} there exists a Nash equilibrium.

Proof. Write $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ for the two pure strategies, and let

$$\Gamma = \{(\mathbf{p}, \mathbf{q}) : p_1, p_2, q_1, q_2 \geq 0,\ p_1 + p_2 = q_1 + q_2 = 1\} \subseteq \mathbb{R}^4,$$

which is (a copy of) the square $[0, 1]^2$, and so closed, bounded and homeomorphic to a disc.

For $(\mathbf{p}, \mathbf{q}) \in \Gamma$ define

$$u_i(\mathbf{p}, \mathbf{q}) = \max\{A(\mathbf{e}_i, \mathbf{q}) - A(\mathbf{p}, \mathbf{q}), 0\}, \qquad v_j(\mathbf{p}, \mathbf{q}) = \max\{B(\mathbf{p}, \mathbf{f}_j) - B(\mathbf{p}, \mathbf{q}), 0\},$$

where $\mathbf{f}_j$ are the pure strategies for Bertha. Thus $u_i > 0$ says 'against $\mathbf{q}$, Albert would do strictly better playing A_i outright'. These are continuous, being maxima of continuous functions. Now set $F(\mathbf{p}, \mathbf{q}) = (\mathbf{p}', \mathbf{q}')$ where

$$p_i' = \frac{p_i + u_i(\mathbf{p}, \mathbf{q})}{1 + u_1(\mathbf{p}, \mathbf{q}) + u_2(\mathbf{p}, \mathbf{q})}, \qquad q_j' = \frac{q_j + v_j(\mathbf{p}, \mathbf{q})}{1 + v_1(\mathbf{p}, \mathbf{q}) + v_2(\mathbf{p}, \mathbf{q})}.$$

The denominators are at least 1, the numerators are non-negative, and $p_1' + p_2' = 1 = q_1' + q_2'$, so $F : \Gamma \to \Gamma$ is a continuous map of Γ into itself. By Brouwer's theorem (in the form of Lemma 3.3) it has a fixed point $(\mathbf{p}^*, \mathbf{q}^*)$.

We claim this is a Nash equilibrium. Write $u_i = u_i(\mathbf{p}^*, \mathbf{q}^*)$. Being a fixed point means $p_i^*(1 + u_1 + u_2) = p_i^* + u_i$, that is

$$p_i^*(u_1 + u_2) = u_i \qquad (i = 1, 2).$$

Since $A(\mathbf{p}^*, \mathbf{q}^*) = p_1^* A(\mathbf{e}_1, \mathbf{q}^*) + p_2^* A(\mathbf{e}_2, \mathbf{q}^*)$ is a convex combination, we cannot have $A(\mathbf{e}_i, \mathbf{q}^*) > A(\mathbf{p}^*, \mathbf{q}^*)$ for both i. So at least one u_i vanishes; without loss of generality $u_2 = 0$. Then $p_2^*(u_1 + u_2) = 0$, so $p_2^* u_1 = 0$. If $u_1 > 0$ then $p_2^* = 0$, so $p_1^* = 1$ and $\mathbf{p}^* = \mathbf{e}_1$, whence $u_1 = \max\{A(\mathbf{e}_1, \mathbf{q}^*) - A(\mathbf{e}_1, \mathbf{q}^*), 0\} = 0$, a contradiction. So $u_1 = u_2 = 0$.

Thus $A(\mathbf{e}_i, \mathbf{q}^*) \leq A(\mathbf{p}^*, \mathbf{q}^*)$ for $i = 1, 2$, and so for any strategy $\mathbf{p}$,

$$A(\mathbf{p}, \mathbf{q}^*) = p_1 A(\mathbf{e}_1, \mathbf{q}^*) + p_2 A(\mathbf{e}_2, \mathbf{q}^*) \leq A(\mathbf{p}^*, \mathbf{q}^*).$$

The same argument applied to v_j gives $B(\mathbf{p}^*, \mathbf{q}) \leq B(\mathbf{p}^*, \mathbf{q}^*)$ for all $\mathbf{q}$. □

Nothing in the argument really used the number 2: with m players having finitely many choices each, exactly the same construction (in a product of simplices) together with Brouwer's theorem in the appropriate dimension gives a stable point. Note also that the IB theory applied only to two players and only to zero-sum games; Nash's theorem is a considerable strengthening.

Two warnings. First, equilibria need not be unique. If Albert and Bertha each choose 'scissors' or 'paper', receiving £1 each if both choose scissors, £2 each if both choose paper, and nothing if they disagree, then both 'everyone plays scissors' and 'everyone plays paper' are stable. Second, and more disturbingly, a unique equilibrium may be very bad for everyone concerned. That, however, is not the mathematician's problem.

Example 4.3 (Chicken)

Albert and Bartholomew drive their cars at one another. If both swerve, each loses 1 prestige point. If exactly one swerves, the swerver loses 5 points and the non-swerver gains 10. If neither swerves, both lose 100.

Let p be the probability that Albert swerves and q the probability that Bartholomew does. Then

$$A(p, q) = -pq - 5p(1-q) + 10(1-p)q - 100(1-p)(1-q) = p(95 - 106q) + 110q - 100,$$

and by symmetry $B(p, q) = q(95 - 106p) + 110p - 100$.

The recipe for problems like this is: first look at the interior of the square, then at the interiors of the edges, then at the vertices. In the interior we need $\partial A/\partial p = 0$ and $\partial B/\partial q = 0$, that is $95 - 106q = 0$ and $95 - 106p = 0$, giving the interior equilibrium

$$(p^*, q^*) = \left(\frac{95}{106}, \frac{95}{106}\right).$$

On the edge $p = 1$ we have $\partial B/\partial q = 95 - 106 = -11 < 0$, so Bartholomew's best response is $q = 0$; and when $q = 0$ we have $\partial A/\partial p = 95 > 0$, so Albert's best response is indeed $p = 1$. Hence $(1, 0)$ is an equilibrium ('Albert always swerves, Bartholomew never does'), and by symmetry so is $(0, 1)$. At $(1, 1)$ we have $\partial A/\partial p = -11 < 0$ and at $(0,0)$ we have $\partial A/\partial p = 95 > 0$, so neither is stable. So there are exactly three Nash equilibria.

It is genuinely easy to solve toy problems in this way; the difficulty is that with m players we must examine the interior of the cube, then each face, then each edge, then each vertex, and the number of steps explodes. So far as anybody knows, this curse of dimensionality is unavoidable.

§5 Dividing the Pot

Faced with the previous section, the tender-hearted ask 'why not cooperate?'. Even granting that people sometimes do, a question remains: how should the gains from cooperation be divided? Nash gave a striking answer.

Definition 5.1 (Convex Set)
A subset E of $\mathbb{R}^m$ is **convex** if $\lambda\mathbf{u} + (1-\lambda)\mathbf{v} \in E$ whenever $\mathbf{u}, \mathbf{v} \in E$ and $0 \leq \lambda \leq 1$.

The set-up is as follows. There are m participants who must jointly choose a point $\mathbf{x}$ of a closed, bounded, convex set $E \subseteq \mathbb{R}^m$; the value of the outcome to the jth participant is x_j. Convexity is not an arbitrary assumption: if the participants can agree on $\mathbf{u}$ or on $\mathbf{v}$, they can agree to toss a suitable coin, and the expected outcome is then $\lambda\mathbf{u}+(1-\lambda)\mathbf{v}$. They also know a **status quo** point $\mathbf{s} \in E$, the outcome if they fail to agree.

Nash argues that a 'best point' $\mathbf{x}^*$ should have the following properties.

(1) $x_j^* \geq s_j$ for all j: nobody is worse off than if they had walked away.

(2) (*Pareto optimality*) If $\mathbf{x} \in E$ and $x_j \geq x_j^*$ for all j, then $\mathbf{x} = \mathbf{x}^*$: there is no available outcome which makes somebody better off and nobody worse off.

(3) (*Independence of irrelevant alternatives*) If $E' \supseteq E$ is closed, bounded and convex and $\mathbf{x}^{**}$ is the best point for $(E', \mathbf{s})$, then $\mathbf{x}^{**} \in E$ implies that $\mathbf{x}^{**}$ is the best point for $(E, \mathbf{s})$.

(4) (*Symmetry*) If E is invariant under permutation of coordinates and $\mathbf{s}$ is too, then $x_1^* = x_2^* = \cdots = x_m^*$.

(5) (*Scale invariance*) If $T\mathbf{x} = (a_1x_1 + b_1, \ldots, a_mx_m + b_m)$ with $a_j > 0$, and $\mathbf{x}^*$ is the best point for $(E, \mathbf{s})$, then $T\mathbf{x}^*$ is the best point for $(TE, T\mathbf{s})$.

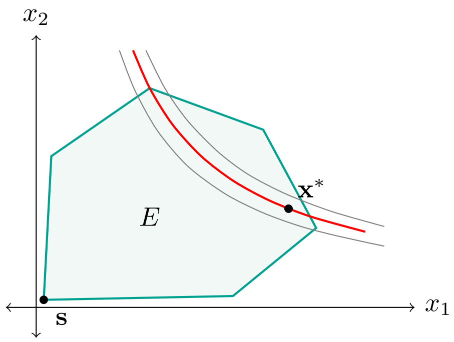
Condition (5) says that we must treat the poor man's penny with the same respect as the rich man's pound: nothing should depend on the units in which each participant measures their satisfaction. It is a genuine assumption, but a reasonable one, and it is the one which does most of the work.

There is no difficulty in remembering these conditions, since each plays a specific role in the proof. We begin with a lemma which is a very concrete instance of the separating hyperplane theorem.

Lemma 5.2
Let K be a convex subset of $\mathbb{R}^n$ with $(1, 1, \ldots, 1) \in K$ and $\prod_{j=1}^n x_j \leq 1$ for all $\mathbf{x} \in K$

with every $x_j \geq 0$. Then
$$K \subseteq \left\{\mathbf{x} : \textstyle\sum_{j=1}^n x_j \leq n\right\}.$$

Proof. Let $\mathbf{x} \in K$ and let $\mathbf{1} = (1, \dots, 1)$. By convexity $\delta\mathbf{x} + (1-\delta)\mathbf{1} \in K$ for $0 \leq \delta \leq 1$, and for δ small enough every coordinate $1 + \delta(x_j - 1)$ is positive. So the hypothesis applies and
$$\prod_{j=1}^n \big(1 + \delta(x_j - 1)\big) \leq 1.$$
Expanding, the left-hand side equals $1 + \delta \sum_{j=1}^n (x_j - 1) + \delta^2 R(\delta)$ where $R(\delta)$ stays bounded as $\delta \to 0$. Hence $\sum_{j=1}^n (x_j - 1) + \delta R(\delta) \leq 0$, and letting $\delta \to 0^+$ gives $\sum_j x_j \leq n$. □



Lemma 5.3

Let $E \subseteq \mathbb{R}^m$ be closed, bounded and convex with $\mathbf{s} \in \operatorname{Int} E$. Then the function $f(\mathbf{x}) = \prod_{j=1}^m (x_j - s_j)$ attains a maximum on $E^+ = \{\mathbf{x} \in E : x_j \geq s_j \text{ for all } j\}$ at exactly one point.

Proof. The set E^+ is closed, bounded and non-empty, so f attains a maximum M there. Since $\mathbf{s} \in \operatorname{Int} E$ we can find $\mathbf{x} \in E$ with $x_j > s_j$ for all j, so $M > 0$.

For uniqueness, suppose $f(\mathbf{x}) = f(\mathbf{y}) = M$ with $\mathbf{x}, \mathbf{y} \in E^+$. Let $\mathbf{z} = \frac{1}{2}(\mathbf{x} + \mathbf{y}) \in E^+$ by convexity. By the AM–GM inequality,
$$z_j - s_j = \frac{(x_j - s_j) + (y_j - s_j)}{2} \geq \sqrt{(x_j - s_j)(y_j - s_j)},$$
with equality if and only if $x_j = y_j$. Multiplying over j (all factors are non-negative) gives $f(\mathbf{z}) \geq \sqrt{f(\mathbf{x})f(\mathbf{y})} = M$. As M is the maximum, we have equality throughout, and since $M > 0$ every factor is strictly positive, so equality in AM–GM forces $x_j = y_j$ for every j. □

Theorem 5.4 (Nash Bargaining)

Suppose that to each problem $(E, \mathbf{s})$ (with $E \subseteq \mathbb{R}^m$ closed, bounded and convex and $\mathbf{s} \in \operatorname{Int} E$) our conditions assign a best point. Then the best point of $(E, \mathbf{s})$ is the unique maximiser of

$$f(\mathbf{x}) = \prod_{j=1}^{m} (x_j - s_j)$$

over $\{\mathbf{x} \in E : x_j \geq s_j\}$.

Proof. Let $\mathbf{x}^*$ be the maximiser given by Lemma 5.3, so $x_j^* > s_j$ for every j. By condition (5) we may apply the affine change of coordinates $x_j \mapsto (x_j - s_j)/(x_j^* - s_j)$; this replaces $\mathbf{s}$ by $\mathbf{0}$ and $\mathbf{x}^*$ by $\mathbf{1} = (1, 1, \ldots, 1)$, and does not change which point is 'best'. So assume $\mathbf{s} = \mathbf{0}$ and $\mathbf{x}^* = \mathbf{1}$.

Maximality of f now reads: $\prod_j x_j \leq 1$ for every $\mathbf{x} \in E$ with all $x_j \geq 0$. As E is convex and contains $\mathbf{1}$, Lemma 5.2 gives $E \subseteq \{\mathbf{x} : \sum_j x_j \leq m\}$. Pick M so large that $E \subseteq [-M, M]^m$ and set

$$\Sigma = \left\{\mathbf{x} \in [-M, M]^m : \textstyle\sum_j x_j \leq m\right\},$$

which is closed, bounded, convex, contains E, contains $\mathbf{0}$ in its interior, and is invariant under permutations of the coordinates.

Consider the problem $(\Sigma, \mathbf{0})$. By symmetry (condition (4)) its best point has all coordinates equal, say $\mathbf{y} = (t, t, \ldots, t)$ with $mt \leq m$, so $t \leq 1$. By Pareto optimality (condition (2)), since $\mathbf{1} \in \Sigma$ dominates $\mathbf{y}$ coordinatewise when $t < 1$, we must have $t = 1$; that is, the best point of $(\Sigma, \mathbf{0})$ is $\mathbf{1} = \mathbf{x}^*$. Finally $\mathbf{x}^* \in E \subseteq \Sigma$, so by independence of irrelevant alternatives (condition (3)) $\mathbf{x}^*$ is the best point of $(E, \mathbf{0})$. □

So the fair division is the one which maximises the *product* of the gains over the status quo – a rule with no obvious a priori justification which nonetheless falls out of five innocuous-looking requirements. There is no patent for immortality under the moon, but I suspect Nash's results will be remembered long after the last celluloid copy of *A Beautiful Mind* has crumbled to dust.

§6 Approximation by Polynomials

It is a guiding idea of both the calculus and of numerical analysis that well behaved functions look like polynomials. Like most guiding principles, it needs to be used judiciously, and this section is devoted to finding out in what sense it is actually true.

If asked to justify the principle we might mutter something about Taylor's theorem. But Cauchy produced the following example to show that this is not enough.

Example 6.1 (Cauchy's Example)
Define $E : \mathbb{R} \to \mathbb{R}$ by

$$E(t) = \begin{cases} \exp(-1/t^2) & t \neq 0, \\ 0 & t = 0. \end{cases}$$

Then E is infinitely differentiable, and $E^{(n)}(0) = 0$ for every n, yet $E(t) \neq 0$ for $t \neq 0$. So the Taylor series of E about 0 converges everywhere, to the wrong function.

Indeed, for $t \neq 0$ an easy induction gives $E^{(n)}(t) = P_n(1/t)E(t)$ for some polynomial P_n: differentiating $P_n(1/t)E(t)$ gives $\left(-t^{-2}P_n'(1/t) + 2t^{-3}P_n(1/t)\right)E(t)$, which is of the same shape. At 0 we induct again: if $E^{(n)}(0) = 0$ then

$$\frac{E^{(n)}(h) - E^{(n)}(0)}{h} = \frac{1}{h}P_n(1/h)\exp(-1/h^2) \to 0$$

as $h \to 0$, since $x^k e^{-x^2} \to 0$ as $|x| \to \infty$ for every k. So $E^{(n+1)}(0)$ exists and is 0.

By Part II this example has been promoted from a counterexample to an invaluable tool: it is what makes partitions of unity and smoothing kernels possible. But it does show that Taylor's theorem is not going to tell us that continuous functions look like polynomials.

§6.1 Interpolation and Chebyshev Polynomials

We might instead mutter something about interpolation. Write $\mathcal{P}_n$ for the vector space of real polynomials of degree at most n.

Lemma 6.2 (Lagrange Interpolation)
Let $x_0, x_1, \ldots, x_n$ be distinct points of $[a, b]$ and let $f : [a, b] \to \mathbb{R}$.

(i) There is at most one $P \in \mathcal{P}_n$ with $P(x_j) = f(x_j)$ for $0 \le j \le n$.

(ii) Writing $e_j(t) = \prod_{k \neq j} \frac{t - x_k}{x_j - x_k}$, the polynomial $P = \sum_{j=0}^n f(x_j)e_j$ lies in $\mathcal{P}_n$ and satisfies $P(x_i) = f(x_i)$ for all i.

(iii) The e_j form a basis of $\mathcal{P}_n$.

Proof. (i) The difference of two such polynomials lies in $\mathcal{P}_n$ and vanishes at $n+1$ points, so is zero by Lemma 1.15. (ii) Each e_j has degree n, and $e_j(x_i) = \delta_{ij}$ by construction, so $P(x_i) = f(x_i)$. (iii) There are $n+1$ of them and $\dim \mathcal{P}_n = n + 1$; they span by (ii). □

So there is always a unique interpolating polynomial, and if f is smooth we can even say exactly how good it is.

Theorem 6.3 (Interpolation Error)
Let $x_0 < x_1 < \cdots < x_n$ be points of $[a, b]$, let $f : [a, b] \to \mathbb{R}$ be $n + 1$ times differentiable, and let $P \in \mathcal{P}_n$ interpolate f at the x_j. Then for each $t \in [a, b]$ there is a $\xi = \xi(t) \in (a, b)$ with

$$f(t) - P(t) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{j=0}^{n} (t - x_j).$$

Proof. If $t = x_i$ for some i both sides vanish, so suppose not, and write $\pi(s) = \prod_{j=0}^n (s - x_j)$, which is a monic polynomial of degree $n + 1$ with $\pi(t) \neq 0$. Choose the constant

$$\lambda = \frac{f(t) - P(t)}{\pi(t)}$$

and set $g(s) = f(s) - P(s) - \lambda\pi(s)$. Then g vanishes at the $n+2$ distinct points $x_0, \dots, x_n, t$. By Rolle's theorem g' vanishes at $n+1$ points strictly between consecutive zeros of g; applying Rolle again, g'' vanishes at n points; and after $n+1$ applications we obtain a $\xi \in (a,b)$ with $g^{(n+1)}(\xi) = 0$. But $P^{(n+1)} = 0$ and $\pi^{(n+1)} = (n+1)!$, so

$$0 = f^{(n+1)}(\xi) - \lambda(n+1)!,$$

which rearranges to the stated formula. □

This is a genuinely informative answer, because it separates the error into two factors of completely different character. The factor $f^{(n+1)}(\xi)/(n+1)!$ depends on f, about which we may know nothing; the factor $\prod_j (t - x_j)$ depends only on *where we chose to interpolate*, and is entirely under our control. So a sensible person picks the x_j to make $\sup_t \left|\prod_j (t - x_j)\right|$ as small as possible. We shall find the optimal choice in Corollary 7.4, and it will turn out not to be the obvious one.

Meanwhile, the theorem should not be read as saying that interpolation always works. It has a hypothesis of $n+1$ derivatives, and even when those exist they may be enormous. Indeed 'interpolates f at $n+1$ points' is in general a long way from 'is close to f'. To see how badly things can go we introduce a family of polynomials which will be with us for the rest of the course.

Theorem 6.4 (Chebyshev Polynomials)
There are polynomials T_n of degree n and U_{n-1} of degree $n-1$ such that

$$T_n(\cos\theta) = \cos n\theta, \qquad U_{n-1}(\cos\theta) = \frac{\sin n\theta}{\sin\theta}$$

for all θ (the second whenever $\sin\theta \neq 0$). The coefficient of t^n in T_n is 2^{n-1} for $n \geq 1$. The roots of T_n are $\cos\left(\frac{(2r+1)\pi}{2n}\right)$ for $0 \leq r \leq n-1$, and the roots of U_{n-1} are $\cos(r\pi/n)$ for $1 \leq r \leq n-1$.

Proof. By de Moivre's theorem and the binomial expansion,

$$\begin{aligned}\cos n\theta + i\sin n\theta &= (\cos\theta + i\sin\theta)^n \\ &= \sum_{0\leq 2r\leq n} (-1)^r \binom{n}{2r} \cos^{n-2r}\theta\, \sin^{2r}\theta \\ &\quad + i \sum_{0\leq 2r+1\leq n} (-1)^r \binom{n}{2r+1} \cos^{n-2r-1}\theta\, \sin^{2r}\theta \cdot \sin\theta.\end{aligned}$$

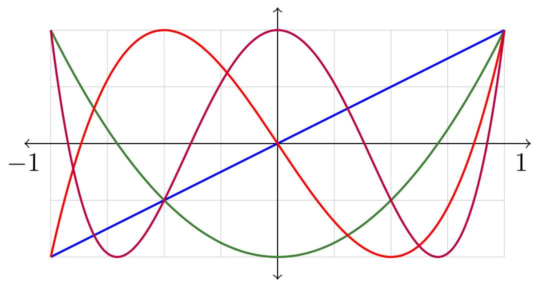
Now $\sin^{2r}\theta = (1-\cos^2\theta)^r$, so comparing real and imaginary parts we get $\cos n\theta = T_n(\cos\theta)$ and $\sin n\theta = \sin\theta\, U_{n-1}(\cos\theta)$ where

$$T_n(t) = \sum_{0\leq 2r\leq n} (-1)^r \binom{n}{2r} t^{n-2r}(1-t^2)^r, \quad U_{n-1}(t) = \sum_{0\leq 2r+1\leq n} (-1)^r \binom{n}{2r+1} t^{n-2r-1}(1-t^2)^r.$$

These are polynomials of degrees n and $n-1$. The coefficient of t^n in T_n comes from taking the top power in each term, giving

$$\sum_{0\le 2r\le n}(-1)^r\binom{n}{2r}(-1)^r=\sum_{0\le 2r\le n}\binom{n}{2r}=\frac{(1+1)^n+(1-1)^n}{2}=2^{n-1}$$

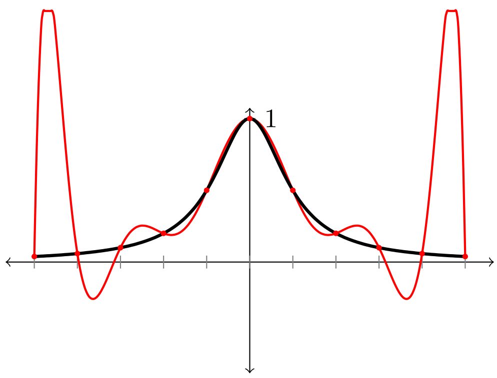
for $n\ge 1$. Finally $T_n(\cos\theta)=0$ exactly when $n\theta$ is an odd multiple of $\pi/2$, and as θ ranges over $[0,\pi]$ this gives the n distinct roots stated; the argument for U_{n-1} is the same. □



The picture already shows the two features of the T_n which will matter to us: on $[-1,1]$ they oscillate between $+1$ and -1, hitting each extreme value alternately and as often as possible, and they are as large as a monic-up-to-scale polynomial can be forced to be while doing so.

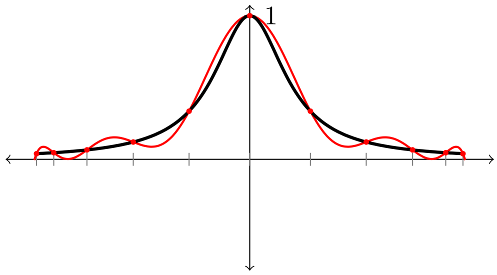
Now look at U_{n-1}. Its roots are the $n-1$ points $\cos(r\pi/n)$, which together with ± 1 are perfectly reasonably spaced points of $[-1,1]$; but $U_{n-1}(1)=n$, which is large. So the function U_{n-1} is 'well behaved' at $n+1$ well spread points (vanishing at $n-1$ of them) and yet is very large elsewhere. Interpolation at innocent-looking points can be badly misleading, and it can be shown – though this is harder – that no choice of interpolation points avoids this in general.

The classical illustration is due to Runge. Take $f(x)=(1+25x^2)^{-1}$ on $[-1,1]$, a function about as well behaved as one could wish, and interpolate it at 11 equally spaced points. Below, the true function is drawn in black and the degree-10 interpolating polynomial in red (clipped where it runs off the picture).



equally spaced nodes

The interpolant is excellent in the middle and disastrous near the endpoints, and the disaster gets worse as the degree increases. If instead we interpolate at the roots of the Chebyshev polynomial T_{11}, which cluster near the endpoints, the same degree gives a genuinely good approximation.



Chebyshev nodes

This is a warning about interpolation, not about polynomials. The question we actually want to answer is whether *some* polynomial is uniformly close to f, and to that we now turn.

§6.2 Which Metric?

Before we can ask whether f looks like g, we must say what we mean. There are several natural metrics on $C([a,b])$, and they are genuinely different.

Example 6.5 (Three Norms)

On $C([0,1])$ the formulae

$$\|f\|_1 = \int_0^1 |f(t)| \,\mathrm{d}t, \qquad \|f\|_2 = \left(\int_0^1 f(t)^2 \,\mathrm{d}t\right)^{1/2}, \qquad \|f\|_\infty = \sup_{t\in[0,1]} |f(t)|$$

all define norms, and hence metrics $d_k(f,g) = \|f-g\|_k$. By the Cauchy–Schwarz inequality $\|f\|_1 \le \|f\|_2$, and clearly $\|f\|_2 \le \|f\|_\infty$, so

$$\|f\|_\infty \ge \|f\|_2 \ge \|f\|_1 .$$

These inequalities go only one way. If $f_n(t) = \max\{1-nt, 0\}$ then $\|f_n\|_\infty = 1$, $\|f_n\|_2 = (3n)^{-1/2}$ and $\|f_n\|_1 = (2n)^{-1}$, so $\|f_n\|_\infty / \|f_n\|_1 = 2n \to \infty$. A tall thin spike is large in $\|\cdot\|_\infty$ and negligible in $\|\cdot\|_1$.

All three are used in practice: $\|\cdot\|_2$ is the natural one when there is an inner product around (as in least squares fitting, and in Fourier analysis), while $\|\cdot\|_1$ is the robust one favoured by statisticians. We shall work almost exclusively with $\|\cdot\|_\infty$, because it is the one for which 'looks like' means what an engineer would want it to mean, and because of the following fact from IB Analysis.

Theorem 6.6

The space $C([a,b])$ with the uniform metric $d(f,g) = \|f-g\|_\infty$ is complete.

This is the general principle of uniform convergence: a uniformly Cauchy sequence of continuous functions converges uniformly, and a uniform limit of continuous functions is continuous.

§6.3 The Weierstrass Approximation Theorem

We can now state the question precisely, and answer it. Weierstrass proved the following in a paper published when he was seventy.

Theorem 6.7 (Weierstrass Approximation Theorem)
Let $f \in C([a,b])$ and $\varepsilon > 0$. Then there is a polynomial P with $\|f - P\|_\infty < \varepsilon$.

Several proofs are known, and it is my belief that one cannot have too many insightful proofs of Weierstrass's theorem. We give one due to Bernstein, which is based on probability and which has the great virtue of writing down an explicit approximating polynomial. We need one tool from IA Probability.

Theorem 6.8 (Chebyshev's Inequality)
If X is a bounded real random variable with $\sigma^2 = \operatorname{Var} X$, then $\Pr(|X - \mathbb{E}X| \geq a) \leq \sigma^2/a^2$ for all $a > 0$.

Proof. Writing $\mu = \mathbb{E}X$ we have $\sigma^2 = \mathbb{E}(X-\mu)^2 \geq \mathbb{E}\left((X-\mu)^2 \mathbb{1}_{|X-\mu|\geq a}\right) \geq a^2 \Pr(|X-\mu| \geq a)$. □

Theorem 6.9 (Bernstein)
Let $f \in C([0,1])$ and define the **Bernstein polynomial**

$$p_n(t) = \sum_{r=0}^{n} \binom{n}{r} f\left(\frac{r}{n}\right) t^r (1-t)^{n-r}.$$

Then $p_n \in \mathcal{P}_n$ and $\|p_n - f\|_\infty \to 0$ as $n \to \infty$.

Proof. The probabilistic idea is this. Fix $t \in [0,1]$ and let $X_1, \dots, X_n$ be independent with $\Pr(X_r = 1) = t$ and $\Pr(X_r = 0) = 1 - t$ – that is, toss a coin with bias t. Put $Y_n = (X_1 + \cdots + X_n)/n$. Then nY_n is binomial, so

$$\mathbb{E}f(Y_n) = \sum_{r=0}^{n} \Pr\left(Y_n = \frac{r}{n}\right) f\left(\frac{r}{n}\right) = \sum_{r=0}^{n} \binom{n}{r} t^r (1-t)^{n-r} f\left(\frac{r}{n}\right) = p_n(t).$$

In other words, $p_n(t)$ is the average of f over a distribution concentrated near t. Since $\mathbb{E}Y_n = t$ and $\operatorname{Var} Y_n = t(1-t)/n \leq 1/(4n)$, the concentration improves as n grows, and continuity of f should do the rest.

Let us do this carefully. Let $\varepsilon > 0$. As $[0,1]$ is compact, f is uniformly continuous (Theorem 1.12), so there is a $\delta > 0$ with $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$. Also $\|f\|_\infty < \infty$. Splitting according to whether Y_n is close to t,

$$\begin{aligned}
|p_n(t) - f(t)| &= \left|\mathbb{E}\big(f(Y_n) - f(t)\big)\right| \\
&\leq \mathbb{E}\left(\mathbb{1}_{|Y_n - t| < \delta} |f(Y_n) - f(t)|\right) + \mathbb{E}\left(\mathbb{1}_{|Y_n - t| \geq \delta} |f(Y_n) - f(t)|\right) \\
&\leq \varepsilon + 2\|f\|_\infty \Pr(|Y_n - t| \geq \delta)
\end{aligned}$$

$$\leq \varepsilon + 2\|f\|_\infty \frac{\operatorname{Var} Y_n}{\delta^2} \leq \varepsilon + \frac{\|f\|_\infty}{2n\delta^2},$$

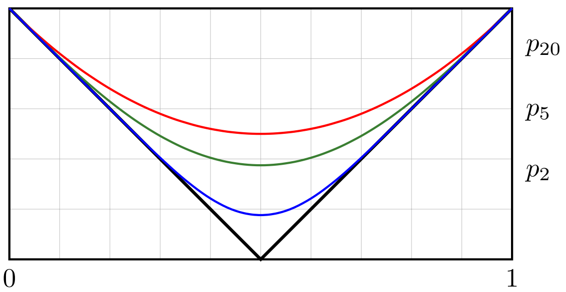
using Chebyshev's inequality. The final bound does not depend on t, so

$$\|p_n - f\|_\infty \leq \varepsilon + \frac{\|f\|_\infty}{2n\delta^2} \leq 2\varepsilon$$

for all n large enough. As $\varepsilon > 0$ was arbitrary, $\|p_n - f\|_\infty \to 0$. □

Proof of Theorem 6.7. Let $f \in C([a,b])$ and set $g(t) = f(a + (b-a)t)$, so $g \in C([0,1])$. By Theorem 6.9 there is a polynomial q with $\|g - q\|_\infty < \varepsilon$, and then $P(x) = q\left(\frac{x-a}{b-a}\right)$ is a polynomial with $\|f - P\|_\infty < \varepsilon$. □

Here is the theorem in action, approximating $f(t) = |t - \frac{1}{2}|$ – a function with a corner, so certainly not analytic – by p_2, p_5 and p_{20}.



Bernstein's polynomials converge, but not quickly: notice that p_{20} is still visibly rounded at the corner. That is the price of an explicit formula which works for every continuous f. It leads naturally to the question of what the *best* polynomial approximation looks like, which is the subject of the next section.

§7 Best Approximation by Polynomials

Bernstein's theorem gives us an explicit approximating polynomial, but except in very special circumstances it is not the *best* one – indeed we have not yet even shown that a best one exists. Chebyshev was very interested in this question, and gave a beautiful criterion for recognising a best approximation when you see one.

The idea is easy to describe. If P is a good approximation to f then the error $f - P$ should not be lopsided: if it were positive more than negative, we could shift P up a little and do better. Chebyshev's insight is that the error of the *best* approximation must swing between its extreme values as often as it possibly can.

Theorem 7.1 (Chebyshev Equiripple Criterion)
Let $f \in C([a,b])$ and $P \in \mathcal{P}_{n-1}$, and write $\sigma = \|f - P\|_\infty$. Suppose we can find points $a \leq a_0 < a_1 < \cdots < a_n \leq b$ and a sign $\zeta \in \{-1, +1\}$ such that

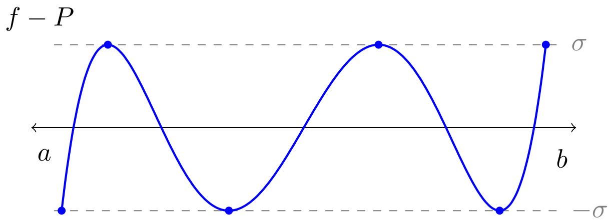
$$f(a_j) - P(a_j) = \zeta(-1)^j \sigma \qquad (0 \leq j \leq n).$$

Then $\|f - P\|_\infty \leq \|f - Q\|_\infty$ for every $Q \in \mathcal{P}_{n-1}$.

Proof. Replacing f and P by $-f$ and $-P$ if necessary, we may assume $\zeta = 1$, so $f(a_j) - P(a_j) = (-1)^j\sigma$. Suppose for contradiction that some $Q \in \mathcal{P}_{n-1}$ has $\|f - Q\|_\infty < \sigma$, and set $R = P - Q \in \mathcal{P}_{n-1}$. Then

$$R(a_j) = \big(f(a_j) - Q(a_j)\big) - \big(f(a_j) - P(a_j)\big) = \big(f(a_j) - Q(a_j)\big) - (-1)^j\sigma.$$

Since $|f(a_j) - Q(a_j)| \le \|f - Q\|_\infty < \sigma$, the sign of $R(a_j)$ is that of $-(-1)^j$, and in particular $R(a_j) \neq 0$. So R takes alternate signs at the $n+1$ points $a_0 < a_1 < \cdots < a_n$, and by the intermediate value theorem R has a root in each of the n intervals (a_j, a_{j+1}). Thus R is a non-zero element of $\mathcal{P}_{n-1}$ with at least n distinct roots, which is impossible. $\square$



Remark. The equiripple condition is also *necessary*: the best approximation always equioscillates in this way, and in particular it is unique. The proof is a pleasant exercise but is not part of the course.

Chebyshev's criterion is designed to be used in one direction: guess a candidate, check that its error equioscillates, and conclude. Let us see the method in action on a concrete problem.

Example 7.2 (The Best Straight Line Through e^t)

We find the best approximation to $f(t) = e^t$ on $[-1, 1]$ from $\mathcal{P}_1$. Write $P(t) = \alpha + \beta t$. Since $n = 2$ here, the criterion asks for three points at which the error takes the values $+\sigma, -\sigma, +\sigma$ (or the negatives).

The error $e^t - \alpha - \beta t$ has derivative $e^t - \beta$, which vanishes exactly once, at $t^* = \log\beta$ (assuming $\beta > 0$). So the error is decreasing then increasing, and its only candidates for extreme values are $t = -1$, $t = t^*$ and $t = 1$. For equioscillation we therefore need

$$e^{-1} - \alpha + \beta \;=\; -\big(e^{t^*} - \alpha - \beta t^*\big) \;=\; e - \alpha - \beta.$$

The outer equality gives $e^{-1} + \beta = e - \beta$, that is

$$\beta = \frac{e - e^{-1}}{2} = \sinh 1 = 1.175201\ldots,$$

which is exactly the slope of the chord of f across $[-1, 1]$ – as one would guess from the mean value theorem. Putting $t^* = \log\beta$ into the first equality and rearranging,

$$\alpha = \frac{e^{-1}}{2} + \beta - \frac{\beta\log\beta}{2} = 1.264300\ldots.$$

The common value of the error is then $\sigma = e^{-1} - \alpha + \beta = 0.278781\ldots$, and by Theorem 7.1 no other line does better. For comparison, the Taylor polynomial

$1 + t$ has error $e - 2 = 0.718\ldots$ at $t = 1$: over an interval of this size, the best approximation is better than the Taylor approximation by a factor of nearly three.

Here is the classic application, and it explains where the Chebyshev polynomials really come from.

Theorem 7.3

Work on $[-1, 1]$ and let $n \geq 1$. Write $S_n(t) = t^n - 2^{1-n}T_n(t)$, which lies in $\mathcal{P}_{n-1}$ by Theorem 6.4. Then

$$\sup_{t\in[-1,1]} |t^n - Q(t)| \geq \sup_{t\in[-1,1]} |t^n - S_n(t)| = 2^{1-n}$$

for every $Q \in \mathcal{P}_{n-1}$.

Proof. By construction $t^n - S_n(t) = 2^{1-n}T_n(t)$, and $|T_n(\cos\theta)| = |\cos n\theta| \leq 1$ with $\|T_n\|_\infty = 1$ on $[-1, 1]$, so $\sigma := \|t^n - S_n\|_\infty = 2^{1-n}$. Let $a_j = \cos\left(\frac{(n-j)\pi}{n}\right)$ for $0 \leq j \leq n$, so that $-1 = a_0 < a_1 < \cdots < a_n = 1$ and

$$a_j^n - S_n(a_j) = 2^{1-n}\cos\big((n-j)\pi\big) = (-1)^{n-j}\sigma = (-1)^n(-1)^j\sigma.$$

So the equiripple criterion of Theorem 7.1 applies with $\zeta = (-1)^n$, and S_n is a best approximation to t^n from $\mathcal{P}_{n-1}$. □

So the Chebyshev polynomial $2^{1-n}T_n$ is precisely the monic polynomial of degree n with smallest uniform norm on $[-1, 1]$. This settles the question we left open when we discussed interpolation.

Corollary 7.4 (Optimal Interpolation Nodes)

Let $x_0, \ldots, x_n$ be distinct points of $[-1, 1]$. Then

$$\sup_{t\in[-1,1]} \left|\prod_{j=0}^{n}(t - x_j)\right| \geq 2^{-n},$$

with equality when the x_j are taken to be the $n + 1$ roots

$$x_j = \cos\left(\frac{(2j+1)\pi}{2n+2}\right) \qquad (0 \leq j \leq n)$$

of T_{n+1}. Consequently, if f is $n + 1$ times differentiable on $[-1, 1]$ and $P \in \mathcal{P}_n$ interpolates f at those roots, then

$$\|f - P\|_\infty \leq \frac{1}{2^n(n+1)!} \sup_{t\in[-1,1]} \left|f^{(n+1)}(t)\right|.$$

Proof. The polynomial $\pi(t) = \prod_{j=0}^{n}(t - x_j)$ is monic of degree $n + 1$, so $\pi(t) = t^{n+1} - Q(t)$ with $Q \in \mathcal{P}_n$, and Theorem 7.3 (with $n+1$ in place of n) gives $\|\pi\|_\infty \geq 2^{1-(n+1)} = 2^{-n}$. Taking the x_j to be the roots of T_{n+1} makes $\pi = 2^{-n}T_{n+1}$, since both sides are monic of degree $n+1$ with the same roots, and $\|T_{n+1}\|_\infty = 1$. The last statement is Theorem 6.3 combined with this equality. $\square$

That is precisely why the second of our two pictures was so much better than the first. For equally spaced nodes the product $\prod_j |t - x_j|$ is small in the middle of the interval but exponentially larger near the ends – which is exactly where the interpolant went wrong – whereas the Chebyshev nodes, which crowd towards ± 1, equalise it. (The reader should not conclude that the corollary *explains* the Runge phenomenon: for $f(x) = (1+25x^2)^{-1}$ the derivatives $f^{(n+1)}$ grow so fast that the bound above is useless for either choice of nodes. What is true is that the bound is the best one available which does not look at f, and that in practice Chebyshev interpolation converges for a much wider class of functions than equally spaced interpolation.)

Theorem 7.3 also gives useful lower bounds on the size of polynomials with large coefficients.

Corollary 7.5

We work on $[-1, 1]$.

(i) If $P(t) = \sum_{j=0}^{n} a_j t^j$ with $|a_n| \geq 1$, then $\|P\|_\infty \geq 2^{1-n}$.

(ii) There is an $\varepsilon(n) > 0$ with the following property: if $P(t) = \sum_{j=0}^{n} a_j t^j$ and $|a_k| \geq 1$ for *some* $0 \leq k \leq n$, then $\|P\|_\infty \geq \varepsilon(n)$.

Proof. (i) Write $P/a_n = t^n - Q(t)$ with $Q \in \mathcal{P}_{n-1}$. By Theorem 7.3, $\|P\|_\infty / |a_n| \geq 2^{1-n}$, and $|a_n| \geq 1$.

(ii) We induct on n. For $n = 0$ we may take $\varepsilon(0) = 1$. Suppose $\varepsilon(n)$ works, and let $P(t) = \sum_{j=0}^{n+1} a_j t^j$ with $|a_k| \geq 1$ for some k. If $|a_{n+1}| \geq \varepsilon(n)/2$ then (i) applied to P/a_{n+1} gives $\|P\|_\infty \geq |a_{n+1}| 2^{-n} \geq \varepsilon(n) 2^{-n-1}$. If instead $|a_{n+1}| < \varepsilon(n)/2$ then $k \leq n$, so writing $R(t) = \sum_{j=0}^{n} a_j t^j$ we have $\|R\|_\infty \geq \varepsilon(n)$ by the inductive hypothesis, and

$$\|P\|_\infty \geq \|R\|_\infty - |a_{n+1}| \left\|t^{n+1}\right\|_\infty \geq \varepsilon(n) - \frac{\varepsilon(n)}{2} = \frac{\varepsilon(n)}{2}.$$

So $\varepsilon(n+1) = \varepsilon(n) 2^{-n-1}$ will do. $\square$

What part (ii) really says is that on the finite dimensional space $\mathcal{P}_n$ the uniform norm and the norm $\max_j |a_j|$ on coefficients are equivalent. That is exactly what we need to run a compactness argument.

Theorem 7.6

If $f \in C([a, b])$ and $n \geq 0$, then there is a $P \in \mathcal{P}_n$ with $\|f - P\|_\infty \leq \|f - Q\|_\infty$ for all $Q \in \mathcal{P}_n$.

Proof. A linear change of variable turns $[a, b]$ into $[-1, 1]$ and carries $\mathcal{P}_n$ bijectively onto itself, so we may assume $[a, b] = [-1, 1]$. Define $\Phi : \mathbb{R}^{n+1} \to \mathbb{R}$ by

$$\Phi(a_0, \dots, a_n) = \left\| f - \sum_{j=0}^{n} a_j t^j \right\|_\infty .$$

Then Φ is continuous: indeed $|\Phi(\mathbf{a}) - \Phi(\mathbf{b})| \le \left\| \sum (a_j - b_j) t^j \right\|_\infty \le \sum_j |a_j - b_j|$.

Let $\kappa = \inf \Phi$, and note $\kappa \le \Phi(\mathbf{0}) = \|f\|_\infty$. If $\Phi(\mathbf{a}) \le \|f\|_\infty$ then $\left\| \sum a_j t^j \right\|_\infty \le 2\|f\|_\infty$, and so, writing $M = \max_j |a_j|$, Corollary 7.5(ii) applied to $M^{-1} \sum a_j t^j$ (which has a coefficient of modulus 1) gives $\left\| \sum a_j t^j \right\|_\infty \ge M\varepsilon(n)$ whenever $M > 0$. Hence

$$M \le \frac{2\|f\|_\infty}{\varepsilon(n)}.$$

So the set $E = \{\mathbf{a} : \Phi(\mathbf{a}) \le \|f\|_\infty\}$ is bounded, and it is closed since Φ is continuous. Thus E is compact and non-empty, so Φ attains a minimum on E; since $\Phi \ge \Phi(\mathbf{0})$ off E, this is a global minimum. □

Remark (Least Squares versus Uniform Approximation)**.** It is worth contrasting this with best approximation in the $\|\cdot\|_2$ norm. There, $\mathcal{P}_n$ is a finite dimensional subspace of an inner product space, and the best approximation is simply the orthogonal projection of f onto $\mathcal{P}_n$: it exists, is unique, is characterised by $\langle f - P, Q \rangle = 0$ for all $Q \in \mathcal{P}_n$, and can be computed by expanding f in an orthonormal basis. All of that is linear algebra. The uniform norm has no inner product behind it, which is why we have had to work for existence, and why the characterisation (equioscillation) is a genuinely analytic statement.

§8 Gaussian Quadrature

Suppose we want to estimate $\int_a^b f(x)\, \mathrm{d}x$ knowing only the values of f at finitely many points. The naive approach is to interpolate and integrate the interpolant, which leads at once to the following.

Lemma 8.1

Let $x_0, \dots, x_n$ be distinct points of $[a, b]$. Then there are unique real numbers $A_0, \dots, A_n$ with

$$\int_a^b P(x)\, \mathrm{d}x = \sum_{j=0}^{n} A_j P(x_j)$$

for all $P \in \mathcal{P}_n$.

Proof. With e_j as in Lemma 6.2, every $P \in \mathcal{P}_n$ satisfies $P = \sum_j P(x_j) e_j$, so setting $A_j = \int_a^b e_j(x)\, \mathrm{d}x$ and integrating gives the formula. Conversely, if the formula holds then putting $P = e_i$ gives $\int_a^b e_i = \sum_j A_j e_i(x_j) = A_i$, so the A_j are determined. □

Given the previous section's warnings about interpolation, we should be suspicious of the resulting approximation $\int_a^b f \approx \sum_j A_j f(x_j)$. Our suspicion is justified: when the x_j are equally spaced (the Newton–Cotes rules), computation shows that as n grows the A_j start to vary in sign and $\sum_j |A_j|$ grows rapidly. Since an error of size η in the values

$f(x_j)$ produces an error of up to $\eta \sum_j |A_j|$ in the answer, the scheme becomes worse as we use more points.

It is a genuinely surprising fact that a clever choice of points avoids this completely. The clever choice is due to Gauss, and involves the orthogonal polynomials for the inner product

$$\langle f, g \rangle = \int_{-1}^{1} f(t)g(t)\, \mathrm{d}t$$

on $C([-1, 1])$.

Definition 8.2 (Legendre Polynomials)

The **Legendre polynomials** $p_0, p_1, p_2, \ldots$ are defined by the conditions that p_n has degree exactly n with positive leading coefficient, and $\langle p_n, p_m \rangle = \delta_{nm}$.

They exist and are unique: apply Gram–Schmidt to $1, t, t^2, \ldots$, so that

$$q_n(t) = t^n - \sum_{m=0}^{n-1} \langle t^n, p_m \rangle p_m(t), \qquad p_n = \frac{q_n}{\|q_n\|_2}.$$

Here $q_n \neq 0$ because it has leading coefficient 1. By construction $p_0, \ldots, p_n$ is an orthonormal basis of $\mathcal{P}_n$, and $p_n \perp \mathcal{P}_{n-1}$. (Other normalisations of the Legendre polynomials are common; they differ from ours only by a positive scalar.)

Lemma 8.3

The polynomial p_n has n distinct roots, all lying in $(-1, 1)$.

Proof. Let $y_1, \ldots, y_m$ be the distinct points of $(-1, 1)$ at which p_n changes sign, and set $Q(t) = \prod_{j=1}^{m}(t - y_j)$ (with $Q = 1$ if $m = 0$). Then $p_n Q$ does not change sign anywhere on $[-1, 1]$, and is not identically zero, so

$$\langle p_n, Q \rangle = \int_{-1}^{1} p_n(t)Q(t)\, \mathrm{d}t \neq 0.$$

But $p_n \perp \mathcal{P}_{n-1}$, so we cannot have $\deg Q = m \leq n - 1$. Hence $m \geq n$; as p_n has degree n it has at most n roots, so $m = n$, the roots are simple and all lie in $(-1, 1)$. $\square$

Theorem 8.4 (Gaussian Quadrature)

Let $\alpha_1, \ldots, \alpha_n$ be the roots of p_n and let $A_1, \ldots, A_n$ be the weights of Lemma 8.1 for these points, so that $\int_{-1}^{1} P = \sum_j A_j P(\alpha_j)$ for all $P \in \mathcal{P}_{n-1}$.

(i) In fact $\int_{-1}^{1} Q(x)\, \mathrm{d}x = \sum_{j=1}^{n} A_j Q(\alpha_j)$ for every $Q \in \mathcal{P}_{2n-1}$.

(ii) Conversely, if $\beta_j \in [-1, 1]$ and B_j are such that $\int_{-1}^{1} Q = \sum_j B_j Q(\beta_j)$ for all $Q \in \mathcal{P}_{2n-1}$, then the β_j are the roots of p_n.

Proof. (i) Let $Q \in \mathcal{P}_{2n-1}$. Dividing by p_n we may write $Q = Gp_n + R$ with $G, R \in \mathcal{P}_{n-1}$. Then

$$\int_{-1}^{1} Q = \int_{-1}^{1} Gp_n + \int_{-1}^{1} R = \langle G, p_n \rangle + \sum_{j=1}^{n} A_j R(\alpha_j) = 0 + \sum_{j=1}^{n} A_j R(\alpha_j),$$

using $p_n \perp \mathcal{P}_{n-1}$ and the defining property of the A_j. But $p_n(\alpha_j) = 0$, so $R(\alpha_j) = Q(\alpha_j)$, giving the result.

(ii) Set $W(x) = \prod_{j=1}^{n}(x - \beta_j)$, a monic polynomial of degree n. For any $R \in \mathcal{P}_{n-1}$ the product WR lies in $\mathcal{P}_{2n-1}$ and vanishes at every β_j, so

$$\langle W, R \rangle = \int_{-1}^{1} WR = \sum_{j=1}^{n} B_j W(\beta_j) R(\beta_j) = 0.$$

Hence $W \in \mathcal{P}_n$ is orthogonal to $\mathcal{P}_{n-1}$. Since $p_0, \dots, p_n$ is an orthonormal basis of $\mathcal{P}_n$ and $p_0, \dots, p_{n-1}$ spans $\mathcal{P}_{n-1}$, the orthogonal complement of $\mathcal{P}_{n-1}$ in $\mathcal{P}_n$ is one-dimensional and spanned by p_n. So W is a scalar multiple of p_n, and the β_j are the roots of p_n. $\square$

Doubling the degree of exactness for free is impressive, but the real payoff is the following, which shows that the objection to Newton–Cotes evaporates.

Theorem 8.5

With the notation of Theorem 8.4, write $G_n f = \sum_{j=1}^{n} A_j f(\alpha_j)$.

(i) $A_j > 0$ for each j.

(ii) $\sum_{j=1}^{n} A_j = 2$.

(iii) If $f \in C([-1,1])$ and $P \in \mathcal{P}_{2n-1}$, then $\left| \int_{-1}^{1} f - G_n f \right| \leq 4 \|f - P\|_\infty$.

(iv) If $f \in C([-1,1])$ then $G_n f \to \int_{-1}^{1} f(t)\, dt$ as $n \to \infty$.

Proof. (i) Fix k and let $w_k(t) = \prod_{j \neq k}(t - \alpha_j)^2$, a polynomial of degree $2n-2 \leq 2n-1$. By Theorem 8.4(i),

$$0 < \int_{-1}^{1} w_k(t)\, dt = \sum_{j=1}^{n} A_j w_k(\alpha_j) = A_k w_k(\alpha_k),$$

since w_k vanishes at every α_j with $j \neq k$. As $w_k(\alpha_k) > 0$ we get $A_k > 0$.

(ii) Take $Q = 1$ in Theorem 8.4(i).

(iii) Since $\int_{-1}^{1} P = G_n P$,

$$\left| \int_{-1}^{1} f - G_n f \right| = \left| \int_{-1}^{1} (f - P) - \sum_j A_j \big(f(\alpha_j) - P(\alpha_j) \big) \right|$$

$$\le \int_{-1}^{1} |f-P| + \sum_j |A_j|\,|f(\alpha_j) - P(\alpha_j)|$$

$$\le 2\,\|f-P\|_\infty + \left(\sum_j A_j\right)\|f-P\|_\infty = 4\,\|f-P\|_\infty\,,$$

where we used (i) to drop the modulus signs on the A_j and (ii) to evaluate their sum.

(iv) Given $\varepsilon > 0$, Weierstrass's Theorem 6.7 provides a polynomial P with $\|f-P\|_\infty < \varepsilon/4$. For all n with $2n-1 \ge \deg P$, part (iii) gives $\left|\int_{-1}^{1} f - G_n f\right| < \varepsilon$. ☐

It is worth carrying the construction out once, if only to see how modest the numbers are.

Example 8.6 (The Two-Point Gauss Rule)

Take $n = 2$. Gram–Schmidt gives $p_0(t) = 1/\sqrt{2}$ and, since $\langle t, p_0\rangle = 0$ and $\int_{-1}^{1} t^2\,\mathrm{d}t = \frac{2}{3}$, also $p_1(t) = \sqrt{3/2}\,t$. Next

$$q_2(t) = t^2 - \langle t^2, p_0\rangle p_0(t) - \langle t^2, p_1\rangle p_1(t) = t^2 - \frac{2/3}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}} - 0 = t^2 - \frac{1}{3},$$

so the nodes are $\alpha_{1,2} = \pm 1/\sqrt{3}$. By symmetry $A_1 = A_2$, and Theorem 8.5(ii) gives $A_1 = A_2 = 1$. Thus

$$\int_{-1}^{1} f(t)\,\mathrm{d}t \approx f\left(-\tfrac{1}{\sqrt{3}}\right) + f\left(\tfrac{1}{\sqrt{3}}\right),$$

and Theorem 8.4 says that this is exact for every cubic. (One can check this by hand: $\int_{-1}^{1} t^2\,\mathrm{d}t = \frac{2}{3} = \frac{1}{3} + \frac{1}{3}$ and $\int_{-1}^{1} t^3\,\mathrm{d}t = 0 = -3^{-3/2} + 3^{-3/2}$.)

Let us try it on $f(t) = e^t$, where $\int_{-1}^{1} e^t\,\mathrm{d}t = e - e^{-1} = 2.350402\ldots$. The rule gives

$$e^{-1/\sqrt{3}} + e^{1/\sqrt{3}} = 2\cosh\left(\tfrac{1}{\sqrt{3}}\right) = 2.342760\ldots,$$

an error of 0.0076 from two evaluations of f. Simpson's rule, which is also exact on cubics, uses *three* evaluations and gives

$$\tfrac{1}{3}\left(e^{-1} + 4e^0 + e\right) = 2.362054\ldots,$$

an error of 0.0117. So Gauss does better with less work, which is the whole point.

It is worth appreciating how the pieces fit together. Positivity of the weights (i) is what stops the errors from amplifying; it is a consequence of the choice of nodes, and it is exactly what fails for equally spaced nodes. Once we have positivity and (ii), the quadrature error is controlled by the best polynomial approximation error, and Weierstrass's theorem then guarantees convergence for *every* continuous f – with no smoothness assumptions at all.

§9 Distance and Compact Sets

This section is a digression, but a useful one: we construct a metric on the collection of compact sets. It provides some helpful background for Runge's theorem, and it will supply a striking application of the Baire category theorem later on.

Throughout we work in $\mathbb{R}^m$ with the usual metric, and all sets are non-empty and compact. Recall from Lemma 1.13 that $d(\mathbf{x}, A) = \inf_{\mathbf{a}\in A} \|\mathbf{x}-\mathbf{a}\|$ defines a continuous function of $\mathbf{x}$.

Lemma 9.1

Let E be a non-empty compact subset of $\mathbb{R}^m$ and $\mathbf{a} \in \mathbb{R}^m$. Then there is a point $\mathbf{e} \in E$ with $\|\mathbf{a}-\mathbf{e}\| = d(\mathbf{a}, E)$.

Proof. The map $\mathbf{x} \mapsto \|\mathbf{a}-\mathbf{x}\|$ is continuous on the compact set E, so attains its infimum by Theorem 1.10. □

The nearest point need not be unique – take E to be a circle and $\mathbf{a}$ its centre – but it is unique if E is convex. Indeed if $\mathbf{u}, \mathbf{v} \in E$ both achieve the minimum distance ρ from $\mathbf{a}$ and $\mathbf{u} \neq \mathbf{v}$, then $\frac{1}{2}(\mathbf{u}+\mathbf{v}) \in E$ and, by the parallelogram law,

$$\left\|\mathbf{a} - \tfrac{\mathbf{u}+\mathbf{v}}{2}\right\|^2 = \rho^2 - \tfrac{1}{4}\|\mathbf{u}-\mathbf{v}\|^2 < \rho^2,$$

a contradiction.

Lemma 9.2

Let $E, F \subseteq \mathbb{R}^m$ be non-empty.

(i) If E and F are compact, there exist $\mathbf{e} \in E$, $\mathbf{f} \in F$ with $\|\mathbf{e}-\mathbf{f}\| \le \|\mathbf{x}-\mathbf{y}\|$ for all $\mathbf{x} \in E$, $\mathbf{y} \in F$.

(ii) The same holds if E is compact and F is closed.

(iii) It may fail if E and F are merely closed.

Proof. (i) The function $(\mathbf{x}, \mathbf{y}) \mapsto \|\mathbf{x}-\mathbf{y}\|$ is continuous on the compact set $E \times F$.

(ii) Choose $R > 0$ with $E \subseteq \overline{B}(\mathbf{0}, R)$ and $F \cap \overline{B}(\mathbf{0}, R) \neq \emptyset$. Set $F_R = F \cap \overline{B}(\mathbf{0}, 3R)$, a non-empty compact set. By (i) there are $\mathbf{e} \in E$, $\mathbf{f} \in F_R$ minimising the distance over $E \times F_R$, and $\|\mathbf{e}-\mathbf{f}\| \le 2R$. If $\mathbf{y} \in F \setminus F_R$ then $\|\mathbf{x}-\mathbf{y}\| \ge 3R - R = 2R$ for $\mathbf{x} \in E$, so the minimum over $E \times F$ is achieved too.

(iii) Take $E = \{(x, 1/x) : x > 0\}$ and $F = \{(-x, 1/x) : x > 0\}$. Both are closed, $\|\mathbf{x}-\mathbf{y}\| > 0$ for all $\mathbf{x} \in E$, $\mathbf{y} \in F$, but taking x large shows the infimum is 0. □

§9.1 The Hausdorff Metric

We would like a notion of distance between compact sets. The first thing one writes down is

$$\tau(E, F) = \inf_{\mathbf{e}\in E} d(\mathbf{e}, F) = \inf\{\|\mathbf{e}-\mathbf{f}\| : \mathbf{e} \in E,\ \mathbf{f} \in F\}.$$

This is symmetric and non-negative, and by Lemma 9.2 we have $\tau(E,F)=0$ if and only if $E\cap F\neq\emptyset$. So it fails the axiom $\tau(E,F)=0\Rightarrow E=F$ rather badly, and it also fails the triangle inequality: with $A=\{1\}$, $B=\{1,2\}$, $C=\{2\}$ we have $\tau(A,C)=1$ but $\tau(A,B)+\tau(B,C)=0$.

The trouble is that an infimum only notices the closest pair of points. Let us try a supremum instead:

$$\sigma(E,F)=\sup_{\mathbf{e}\in E} d(\mathbf{e},F).$$

This is non-negative, and $\sigma(E,F)=0$ if and only if $d(\mathbf{e},F)=0$ for every $\mathbf{e}\in E$, that is (as F is closed) if and only if $E\subseteq F$. So σ measures 'how far E sticks out of F', and is emphatically not symmetric. But it does obey the triangle inequality.

Lemma 9.3

For non-empty compact E,F,G we have $\sigma(E,G)\leq\sigma(E,F)+\sigma(F,G)$.

Proof. Let $\mathbf{e}\in E$. By Lemma 9.1 choose $\mathbf{f}\in F$ with $\|\mathbf{e}-\mathbf{f}\|=d(\mathbf{e},F)$. For any $\mathbf{g}\in G$,

$$d(\mathbf{e},G)\leq\|\mathbf{e}-\mathbf{g}\|\leq\|\mathbf{e}-\mathbf{f}\|+\|\mathbf{f}-\mathbf{g}\|=d(\mathbf{e},F)+\|\mathbf{f}-\mathbf{g}\|.$$

Taking the infimum over $\mathbf{g}\in G$ gives $d(\mathbf{e},G)\leq d(\mathbf{e},F)+d(\mathbf{f},G)\leq\sigma(E,F)+\sigma(F,G)$, and taking the supremum over $\mathbf{e}\in E$ finishes the proof. $\square$

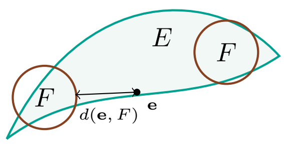
Symmetrising now gives what we want.

Definition 9.4 (Hausdorff Metric)

Let $\mathcal{K}$ be the collection of non-empty compact subsets of $\mathbb{R}^m$. For $E,F\in\mathcal{K}$ set

$$\rho(E,F)=\sigma(E,F)+\sigma(F,E)=\sup_{\mathbf{e}\in E}\inf_{\mathbf{f}\in F}\|\mathbf{e}-\mathbf{f}\|+\sup_{\mathbf{f}\in F}\inf_{\mathbf{e}\in E}\|\mathbf{e}-\mathbf{f}\|.$$

We call ρ the **Hausdorff metric**.



Theorem 9.5

ρ is a metric on $\mathcal{K}$.

Proof. Clearly $\rho\geq 0$ and ρ is symmetric. If $\rho(E,F)=0$ then $\sigma(E,F)=\sigma(F,E)=0$, so $E\subseteq F$ and $F\subseteq E$, giving $E=F$; and conversely. The triangle inequality follows from Lemma 9.3 applied twice:

$$\rho(E,G)=\sigma(E,G)+\sigma(G,E)\leq\big(\sigma(E,F)+\sigma(F,G)\big)+\big(\sigma(G,F)+\sigma(F,E)\big)=\rho(E,F)+\rho(F,G).$$

$\square$

What makes ρ genuinely useful is that it is complete. We need two preliminary observations.

Lemma 9.6

Suppose $K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$ are non-empty compact subsets of $\mathbb{R}^m$ and let $K = \bigcap_{p\geq 1} K_p$. Then K is non-empty and compact, and $\rho(K_p, K) \to 0$.

Proof. K is closed (an intersection of closed sets) and bounded, hence compact. To see it is non-empty, pick $\mathbf{k}_p \in K_p$ for each p. All the $\mathbf{k}_p$ lie in the compact set K_1, so some subsequence $\mathbf{k}_{p(j)} \to \mathbf{k}$. For each fixed q we have $\mathbf{k}_{p(j)} \in K_q$ for all j with $p(j) \geq q$, and K_q is closed, so $\mathbf{k} \in K_q$. Hence $\mathbf{k} \in K$.

Since $K \subseteq K_p$ we have $\sigma(K, K_p) = 0$, so $\rho(K_p, K) = \sigma(K_p, K)$. Suppose this does not tend to 0. Then there is an $\varepsilon > 0$, a subsequence $p(j)$, and points $\mathbf{x}_{p(j)} \in K_{p(j)}$ with $d(\mathbf{x}_{p(j)}, K) \geq \varepsilon$. Passing to a further subsequence, $\mathbf{x}_{p(j)} \to \mathbf{x}$, and exactly as above $\mathbf{x} \in K$. But $d(\cdot, K)$ is continuous, so $d(\mathbf{x}, K) \geq \varepsilon > 0$, contradicting $\mathbf{x} \in K$. □

Lemma 9.7

If $K \subseteq \mathbb{R}^m$ is compact and $r \geq 0$, then $K + \overline{B}(\mathbf{0}, r) = \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in K, \|\mathbf{y}\| \leq r\}$ is compact.

Proof. It is clearly bounded. If $\mathbf{x}_n + \mathbf{y}_n \to \mathbf{c}$ with $\mathbf{x}_n \in K$ and $\|\mathbf{y}_n\| \leq r$, pass to a subsequence with $\mathbf{x}_{n(j)} \to \mathbf{x} \in K$; then $\mathbf{y}_{n(j)} \to \mathbf{c} - \mathbf{x}$, so $\|\mathbf{c} - \mathbf{x}\| \leq r$ and $\mathbf{c} \in K + \overline{B}(\mathbf{0}, r)$. □

Theorem 9.8

The Hausdorff metric ρ is complete on $\mathcal{K}$.

Proof. By Lemma 1.3(ii) it suffices to show that any sequence (E_n) in $\mathcal{K}$ with $\rho(E_n, E_{n+1}) \leq 100^{-n}$ converges. Set

$$K_n = E_n + \overline{B}(\mathbf{0}, 10 \cdot 100^{-n}),$$

which is compact by Lemma 9.7. Since $\sigma(E_{n+1}, E_n) \leq \rho(E_n, E_{n+1}) \leq 100^{-n}$, every point of E_{n+1} lies within 100^{-n} of E_n, that is $E_{n+1} \subseteq E_n + \overline{B}(\mathbf{0}, 100^{-n})$. Hence

$$K_{n+1} \subseteq E_n + \overline{B}(\mathbf{0}, 100^{-n}) + \overline{B}(\mathbf{0}, 10 \cdot 100^{-n-1}) \subseteq E_n + \overline{B}(\mathbf{0}, 2 \cdot 100^{-n}) \subseteq K_n.$$

By Lemma 9.6 the set $K = \bigcap_n K_n$ is a non-empty compact set with $\rho(K_n, K) \to 0$. Finally $E_n \subseteq K_n$ and every point of K_n is within $10 \cdot 100^{-n}$ of E_n, so $\rho(E_n, K_n) \leq 10 \cdot 100^{-n} \to 0$, and therefore

$$\rho(E_n, K) \leq \rho(E_n, K_n) + \rho(K_n, K) \to 0.$$ □

§10 Runge's Theorem

Weierstrass's theorem tells us that every continuous real function on $[a, b]$ is a uniform limit of polynomials. Is there a complex analogue? Cauchy's theorem answers with a resounding no.

Example 10.1

Let $\overline{D} = \{z : |z| \le 1\}$ and $f(z) = \overline{z}$. Then $\sup_{z\in\overline{D}} |f(z) - P(z)| \ge 1$ for every polynomial P.

Indeed, let C be the contour $\theta \mapsto e^{i\theta}$, $0 \le \theta \le 2\pi$. By Cauchy's theorem $\int_C P(z)\,\mathrm{d}z = 0$, so

$$2\pi \left\|f - P\right\|_\infty \ge \left|\int_C (\overline{z} - P(z))\,\mathrm{d}z\right| = \left|\int_C \overline{z}\,\mathrm{d}z\right| = \left|\int_0^{2\pi} e^{-i\theta}\cdot ie^{i\theta}\,\mathrm{d}\theta\right| = 2\pi.$$

That example is unfair: $\overline{z}$ is nowhere analytic, and we know that a uniform limit of analytic functions is analytic. So let us ask for analytic f. Cauchy's theorem still causes trouble.

Example 10.2

Let $T = \{z : \frac{1}{2} \le |z| \le 2\}$ and $f(z) = 1/z$, which is analytic on the open set $\Omega = \{z : 0 < |z| < 3\} \supseteq T$. Then $\sup_{z\in T} |f(z) - P(z)| \ge 1$ for every polynomial P, since with C as above,

$$2\pi \left\|f - P\right\|_\infty \ge \left|\int_C \left(\frac{1}{z} - P(z)\right)\,\mathrm{d}z\right| = |2\pi i| = 2\pi.$$

The obstruction here is that T has a hole, and f misbehaves inside it. So the best we can hope for is a theorem about sets without holes, and the following turns out to be the right notion.

Definition 10.3 (Path-Connectedness)

A set $U \subseteq \mathbb{C}$ is **path-connected** if, given $z_0, z_1 \in U$, we can find a continuous $\gamma : [0,1] \to U$ with $\gamma(0) = z_0$ and $\gamma(1) = z_1$.

Theorem 10.4 (Runge)

Let $\Omega \subseteq \mathbb{C}$ be open and $f : \Omega \to \mathbb{C}$ analytic. Suppose $K \subseteq \Omega$ is compact with $\mathbb{C} \setminus K$ path-connected. Then given $\varepsilon > 0$ there is a polynomial P with

$$\sup_{z\in K} |f(z) - P(z)| < \varepsilon.$$

Note that this is genuinely stronger than Taylor's theorem, which says only that f is uniformly approximable by polynomials on any *disc* contained in Ω; Runge's theorem is a global statement. (And there is no hope of simply expanding in a Taylor series: on $\Omega = \{z : 10^{-2} < |z| < 1\} \setminus (-\infty, 0]$ the function $1/z$ is analytic and bounded but is not given by any single power series.)

The proof falls into three steps. First we express f on K as a sum of contour integrals along segments lying outside K; then we approximate each such integral by a finite sum of simple poles; then we show that each simple pole can be pushed off to infinity and replaced by a polynomial. The last step is where path-connectedness enters.

§10.1 Step One: A Contour Representation

Lemma 10.5

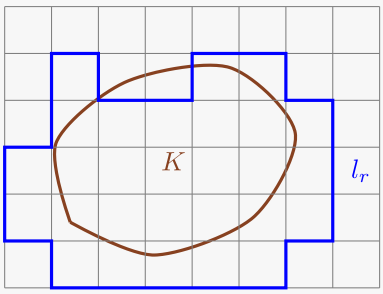
Let $\Omega \subseteq \mathbb{C}$ be open and $K \subseteq \Omega$ compact and non-empty. Then we can find finitely many line segments $l_1, \dots, l_M$, each contained in $\Omega \setminus K$, such that

$$f(z) = \sum_{r=1}^{M} \frac{1}{2\pi i} \int_{l_r} \frac{f(w)}{w - z} \, \mathrm{d}w$$

for every $z \in K$ and every analytic $f : \Omega \to \mathbb{C}$. Moreover $d(K, l_1 \cup \dots \cup l_M) > 0$.

Proof. Since K is compact and $\mathbb{C} \setminus \Omega$ is closed and disjoint from K, Lemma 9.2(ii) gives a $\delta > 0$ with $|k - u| \geq 100\delta$ for all $k \in K$ and $u \notin \Omega$ (if $\Omega = \mathbb{C}$, take any $\delta > 0$).

Tile the plane by the closed squares of side δ with corners in $\delta\mathbb{Z} + i\delta\mathbb{Z}$, and let Γ be the (finite) collection of those squares which meet K. Every $S \in \Gamma$ satisfies $S \subseteq K + \overline{B}(0, 2\delta) \subseteq \Omega$, so f is analytic on a neighbourhood of each $S \in \Gamma$. Write $G = \bigcup_{S \in \Gamma} S$. If $z \in K$ then every square containing z meets K and so lies in Γ; as the squares containing z cover a neighbourhood of z, we get $K \subseteq \operatorname{Int} G$.



Now fix $z \in \operatorname{Int} G$ which does not lie on the boundary of any square of the tiling. Then z lies in the interior of exactly one square S_0, and $S_0 \in \Gamma$ since $z \in G$. Orienting each ∂S anticlockwise, Cauchy's integral formula and Cauchy's theorem give

$$\frac{1}{2\pi i} \int_{\partial S} \frac{f(w)}{w - z} \, \mathrm{d}w = \begin{cases} f(z) & S = S_0, \\ 0 & S \in \Gamma, \ S \neq S_0, \end{cases}$$

so summing over $S \in \Gamma$ yields $f(z) = \sum_{S \in \Gamma} \frac{1}{2\pi i} \int_{\partial S} \frac{f(w)}{w - z} \, \mathrm{d}w$.

Each edge of the tiling which is shared by two squares of Γ is traversed once in each direction, so those contributions cancel. Let $l_1, \dots, l_M$ be the remaining edges – the edges of squares of Γ whose neighbouring square is not in Γ – with their induced orientations. Then

$$f(z) = \sum_{r=1}^{M} \frac{1}{2\pi i} \int_{l_r} \frac{f(w)}{w - z} \, \mathrm{d}w$$

for all such z. Each l_r lies in Ω, and $l_r \cap K = \emptyset$: if l_r met K then the square on its far side would meet K too and would belong to Γ, so l_r would have cancelled. Since $\bigcup_r l_r$ and K are disjoint compact sets, Lemma 9.2(i) gives $d(K, \bigcup_r l_r) > 0$.

Finally, both sides of the displayed identity are continuous functions of z on the open set $\operatorname{Int} G$ (for the right-hand side, because the integrands have denominators bounded away from zero there). They agree on $\operatorname{Int} G$ minus the union of the grid lines, which is dense in the open set $\operatorname{Int} G$. Hence they agree on all of $\operatorname{Int} G \supseteq K$. $\square$

§10.2 Step Two: Approximation by Simple Poles

Lemma 10.6

Let Ω be open, $K \subseteq \Omega$ compact, and $f : \Omega \to \mathbb{C}$ analytic. Given $\varepsilon > 0$ we can find N, complex numbers $A_1, \dots, A_N$ and points $\alpha_1, \dots, \alpha_N \in \Omega \setminus K$ such that

$$\left| f(z) - \sum_{n=1}^{N} \frac{A_n}{z - \alpha_n} \right| \le \varepsilon$$

for all $z \in K$.

Proof. By Lemma 10.5 it suffices to approximate a single term $\frac{1}{2\pi i} \int_l \frac{f(w)}{w-z}\, \mathrm{d}w$, where l is a segment disjoint from K and contained in Ω. Parametrise l by $\gamma(t) = w_0 + t(w_1 - w_0)$ for $t \in [0,1]$, so that

$$\frac{1}{2\pi i} \int_l \frac{f(w)}{w-z}\, \mathrm{d}w = \int_0^1 F(t,z)\, \mathrm{d}t, \qquad F(t,z) = \frac{1}{2\pi i} \cdot \frac{f(\gamma(t))(w_1 - w_0)}{\gamma(t) - z}.$$

Since $d(K, l) > 0$, the function F is continuous on the compact set $[0,1] \times K$, hence uniformly continuous there by Theorem 1.12. So given $\eta > 0$ there is an m with $|F(t,z) - F(r/m, z)| < \eta$ whenever $t \in [r/m, (r+1)/m]$ and $z \in K$. Therefore

$$\left| \int_0^1 F(t,z)\, \mathrm{d}t - \frac{1}{m} \sum_{r=0}^{m-1} F(r/m, z) \right| \le \eta$$

for all $z \in K$, and the Riemann sum is exactly of the form $\sum_r \frac{A_r}{z - \alpha_r}$ with $\alpha_r = \gamma(r/m) \in l \subseteq \Omega \setminus K$ and suitable constants A_r. Doing this for each of the M segments with $\eta = \varepsilon/M$ completes the proof. $\square$

§10.3 Step Three: Pushing Poles to Infinity

Everything now reduces to the following statement, which is where the topology of K finally matters.

Lemma 10.7

Let $K \subseteq \mathbb{C}$ be compact with $\mathbb{C} \setminus K$ path-connected. Then for every $\alpha \notin K$ and every $\varepsilon > 0$ there is a polynomial P with $\left| P(z) - \frac{1}{z-\alpha} \right| < \varepsilon$ for all $z \in K$.

Write $\Lambda(K)$ for the set of $\alpha \notin K$ with this property. We must show $\Lambda(K) = \mathbb{C} \setminus K$, and we do it in three moves.

Proof. **(a) $\Lambda(K)$ contains all sufficiently large α.** Choose R with $K \subseteq B(0, R/2)$.

If $|\alpha| \geq R$ then for $z \in K$,

$$\frac{1}{z-\alpha} = \frac{-1}{\alpha(1-z/\alpha)} = -\frac{1}{\alpha}\sum_{n=0}^{\infty}\left(\frac{z}{\alpha}\right)^n,$$

and since $|z/\alpha| \leq \frac{1}{2}$ on K the series converges uniformly on K by the Weierstrass M-test. Its partial sums are polynomials, so $\alpha \in \Lambda(K)$.

(b) $\Lambda(K)$ is 'locally closed under small moves'. Suppose $\alpha \in \Lambda(K)$ and $|\alpha-\beta| < \frac{1}{2}d(\alpha, K)$. Put $\delta = d(\alpha, K)$, so $|z-\alpha| \geq \delta$ for $z \in K$. Then for $z \in K$,

$$\frac{1}{z-\beta} = \frac{1}{(z-\alpha)-(\beta-\alpha)} = \frac{1}{z-\alpha}\cdot\frac{1}{1-\frac{\beta-\alpha}{z-\alpha}} = \sum_{n=0}^{\infty}\frac{(\beta-\alpha)^n}{(z-\alpha)^{n+1}},$$

and since $\left|\frac{\beta-\alpha}{z-\alpha}\right| \leq \frac{1}{2}$ on K the series converges uniformly on K. So given $\varepsilon > 0$ we can find N with

$$\left|\frac{1}{z-\beta} - \sum_{n=0}^{N}\frac{(\beta-\alpha)^n}{(z-\alpha)^{n+1}}\right| < \frac{\varepsilon}{2} \qquad (z \in K).$$

Now $\alpha \in \Lambda(K)$ means $(z-\alpha)^{-1}$ is uniformly approximable on K by polynomials; hence so is each power $(z-\alpha)^{-(n+1)}$ (a product of uniformly bounded uniformly approximable functions is uniformly approximable), and hence so is the finite sum. Choosing a polynomial within $\varepsilon/2$ of it gives $\beta \in \Lambda(K)$.

(c) Path-connectedness. Let $\beta \notin K$. By (a) choose α_0 with $|\alpha_0|$ large enough that $\alpha_0 \in \Lambda(K)$. Since $\mathbb{C} \setminus K$ is path-connected there is a continuous $\gamma : [0,1] \to \mathbb{C} \setminus K$ with $\gamma(0) = \alpha_0$, $\gamma(1) = \beta$. The image $\gamma([0,1])$ is compact and disjoint from the compact set K, so by Lemma 9.2(i) there is a $\delta > 0$ with $d(\gamma(t), K) \geq 4\delta$ for all t. Also γ is uniformly continuous, so there is an N with $|\gamma(s) - \gamma(t)| < \delta$ whenever $|s-t| \leq 1/N$. Setting $w_j = \gamma(j/N)$ we have $|w_j - w_{j+1}| < \delta \leq \frac{1}{2}d(w_j, K)$, so by (b) and induction $w_0 \in \Lambda(K)$ gives $w_N = \beta \in \Lambda(K)$. □

Proof of Runge's theorem. Given $\varepsilon > 0$, Lemma 10.6 provides A_n and $\alpha_n \notin K$ with $\left|f(z) - \sum_{n=1}^{N}\frac{A_n}{z-\alpha_n}\right| \leq \varepsilon/2$ on K. By Lemma 10.7 each $\frac{1}{z-\alpha_n}$ is within $\frac{\varepsilon}{2N(1+|A_n|)}$ of a polynomial P_n uniformly on K. Then $P = \sum_n A_n P_n$ satisfies $\sup_K |f-P| < \varepsilon$. □

§10.4 An Application: Pointwise Limits Need Not Be Continuous

If analytic functions converge *uniformly*, the limit is analytic. It is natural to ask whether uniformity is really needed, and Runge's theorem lets us answer emphatically.

Example 10.8

Let $D = \{z : |z| < 1\}$ and define $f : \mathbb{C} \to \mathbb{C}$ by $f(0) = 0$ and

$$f(re^{i\theta}) = r^{3/2}e^{3i\theta/2} \qquad (r > 0,\ 0 < \theta \leq 2\pi).$$

Then f is not continuous on D (approaching a positive real x from above gives $x^{3/2}$, from below gives $-x^{3/2}$), yet there are polynomials P_n with $P_n(z) \to f(z)$ for every

$z \in D$.

Proof. The function f is analytic on $\Omega = \mathbb{C} \setminus [0, \infty)$, being a branch of $z^{3/2}$ there. For $n \geq 1$ let $f_n(z) = f(z - i/n)$, which is analytic on $\Omega_n = \mathbb{C} \setminus \left(\frac{i}{n} + [0, \infty)\right)$, and let

$$K_n = \left\{z : |z| \leq 1,\ d\left(z, \tfrac{i}{n} + [0, \infty)\right) \geq \tfrac{1}{n^2}\right\}.$$

Then K_n is compact, $K_n \subseteq \Omega_n$, and $\mathbb{C} \setminus K_n$ is path-connected: K_n is a closed disc with a slit-shaped neighbourhood removed, and any point of the complement can be joined to infinity by going along the slit and out. By Runge's theorem there is a polynomial P_n with $\sup_{z \in K_n} |f_n(z) - P_n(z)| \leq 2^{-n}$.

Now fix z with $|z| < 1$. If z is a non-negative real then $d(z, \frac{i}{n} + [0, \infty)) = \frac{1}{n} \geq \frac{1}{n^2}$, so $z \in K_n$ for every n. Otherwise $d(z, [0, \infty)) = c > 0$ and so $d(z, \frac{i}{n} + [0, \infty)) \geq c - \frac{1}{n} \geq \frac{1}{n^2}$ for all large n. Either way $z \in K_n$ eventually, and then

$$|P_n(z) - f(z)| \leq 2^{-n} + |f(z - i/n) - f(z)| \to 0,$$

since $z - i/n \to z$ from below, and f *is* continuous from below across the positive real axis (as $\theta \to 2\pi^-$ we get $r^{3/2} e^{3i\pi} = -r^{3/2}$, which is the assigned value at $\theta = 2\pi$); at $z = 0$ we simply have $f(-i/n) = n^{-3/2} e^{9i\pi/4} \to 0 = f(0)$. $\square$

§11 Odd Numbers

According to Von Neumann, 'in mathematics you don't understand things, you just get used to them'. The real line is a case in point. A single innocuous-looking axiom – every bounded increasing sequence converges – calls into being an object of indescribable complexity. We know that $\mathbb{R}$ is uncountable; on the other hand, if our alphabet has n symbols then there are at most n^m phrases of length exactly m, so the collection of *describable* real numbers is a countable union of finite sets, and hence countable. Almost every real number cannot be described at all.

In this section we look at a few of the prettier shells: numbers we can describe, and about which we can say something.

§11.1 Irrationality

Theorem 11.1

The number e is irrational.

Proof. Suppose $e = p/q$ with $p, q \in \mathbb{N}$. Recall $e = \sum_{r=0}^{\infty} \frac{1}{r!}$. Consider

$$N = q! \left(e - \sum_{r=0}^{q} \frac{1}{r!}\right) = q!\, e - \sum_{r=0}^{q} \frac{q!}{r!}.$$

Now $q!\, e = q!\, p/q = (q-1)!\, p$ is an integer, and each $q!/r!$ with $r \leq q$ is an integer, so $N \in \mathbb{Z}$. On the other hand

$$0 < N = q! \sum_{r=q+1}^{\infty} \frac{1}{r!} = \sum_{k=1}^{\infty} \frac{q!}{(q+k)!} \leq \sum_{k=1}^{\infty} \frac{1}{(q+1)^k} = \frac{1}{q} \leq 1,$$

and the inequality $q!/(q+k)! \leq (q+1)^{-k}$ is strict for $k \geq 2$, so $0 < N < 1$. There is no such integer. $\square$

The proof of the corresponding statement for π is harder, and rests on the following computation.

Lemma 11.2

Write $f_n(x) = x^n(\pi - x)^n$. Then

$$\int_0^\pi f_n(x) \sin x \,\mathrm{d}x = n! \sum_{j=0}^{n} a_j \pi^j$$

for some integers a_j (depending on n).

Proof. Expanding, $f_n(x) = \sum_{r=0}^{n} \binom{n}{r}(-1)^r \pi^{n-r} x^{n+r}$, a polynomial with terms of degrees between n and $2n$. So $f_n^{(k)}(0) = 0$ for $k < n$ and for $k > 2n$, while for $n \leq k \leq 2n$ only the term with $n + r = k$ survives and

$$f_n^{(k)}(0) = k! \binom{n}{k-n} (-1)^{k-n} \pi^{2n-k}.$$

Since $n \leq k$, the number $k!/n!$ is an integer, so $f_n^{(k)}(0) = n!\, m_k \pi^{2n-k}$ for some $m_k \in \mathbb{Z}$, and the exponent $2n - k$ lies between 0 and n. Moreover $f_n(\pi - x) = f_n(x)$, so $f_n^{(k)}(\pi) = (-1)^k f_n^{(k)}(0)$ is of the same shape.

Now integrate by parts twice. Writing $I_k = \int_0^\pi f_n^{(k)}(x) \sin x \,\mathrm{d}x$,

$$I_k = \Big[-f_n^{(k)}(x) \cos x \Big]_0^\pi + \int_0^\pi f_n^{(k+1)}(x) \cos x \,\mathrm{d}x = f_n^{(k)}(\pi) + f_n^{(k)}(0) + \int_0^\pi f_n^{(k+1)} \cos x \,\mathrm{d}x,$$

and $\int_0^\pi f_n^{(k+1)} \cos x \,\mathrm{d}x = \Big[f_n^{(k+1)} \sin x \Big]_0^\pi - I_{k+2} = -I_{k+2}$. Hence

$$I_k = f_n^{(k)}(\pi) + f_n^{(k)}(0) - I_{k+2}.$$

Since $f_n^{(k)} \equiv 0$ for $k > 2n$, we have $I_k = 0$ for $k > 2n$, and iterating the recursion downwards from $k = 0$ expresses $I_0 = \int_0^\pi f_n \sin x \,\mathrm{d}x$ as an integer combination of the numbers $f_n^{(k)}(0)$ and $f_n^{(k)}(\pi)$. Each of these is $n!$ times an integer times a power π^j with $0 \leq j \leq n$, which is the claim. $\square$

Theorem 11.3

The number π is irrational.

Proof. Write $P_n(X) = \sum_{j=0}^n a_j X^j \in \mathbb{Z}[X]$ as in Lemma 11.2, so that $\int_0^\pi f_n(x) \sin x \, \mathrm{d}x = n!\, P_n(\pi)$. On $(0, \pi)$ we have $x^n(\pi - x)^n \sin x > 0$, so this integral is strictly positive. On the other hand, by AM–GM, $x(\pi - x) \le (\pi/2)^2$, so

$$0 < n!\, P_n(\pi) = \int_0^\pi f_n(x) \sin x \, \mathrm{d}x \le \int_0^\pi \left(\frac{\pi}{2}\right)^{2n} \mathrm{d}x = \pi \left(\frac{\pi^2}{4}\right)^n.$$

Suppose now $\pi = p/q$ with $p, q \in \mathbb{N}$. Since $\deg P_n \le n$, the number $q^n P_n(p/q)$ is an integer, and it is strictly positive, so $q^n P_n(\pi) \ge 1$. Hence

$$n! \le n!\, q^n P_n(\pi) \le \pi \left(\frac{q\pi^2}{4}\right)^n.$$

But $C^n/n! \to 0$ for every constant C, so this fails for large n. □

Faced with a proof like that, the reader may cry 'how did you think of looking at f_n?'. The honest answer is 'I did not, I learnt it from somebody else'; but even granting that we could not have invented it in a thousand years, once presented with it we can see a path which might have led there. We are all familiar with the fact that $\int_0^\pi x^n \sin x \, \mathrm{d}x$ takes the form $U(\pi)$ for a polynomial U of degree at most n with integer coefficients, and hence so does $\int_0^\pi Q(x) \sin x \, \mathrm{d}x$ for any $Q \in \mathbb{Z}[X]$. If $\pi = p/q$ then $q^n U(\pi)$ is an integer. So we experiment, trying to squeeze $\int_0^\pi Q(x) \sin x \, \mathrm{d}x$ between 0 and 1 in the manner of our proof for e; the factor $n!$ appears because we need the integral to be very small, and the symmetric choice $x^n(\pi - x)^n$ is the natural one.

It is worth remembering that after three hundred years we still do not know whether Euler's constant

$$\gamma = \lim_{N \to \infty} \left(\sum_{n=1}^N \frac{1}{n} - \log N \right)$$

is irrational.

§11.2 Transcendence

If the rationals are the best understood numbers, the algebraic numbers are the next best understood.

Definition 11.4 (Algebraic and Transcendental)

A real number α is **algebraic** if it is a root of some non-zero polynomial with integer coefficients. Otherwise α is **transcendental**.

Multiplying up denominators, α is algebraic if and only if it is a root of a non-zero polynomial with rational coefficients.

Lemma 11.5

The set of algebraic numbers is countable. Consequently transcendental numbers exist, and indeed the transcendental numbers are uncountable.

Proof. For each n and each bound B there are finitely many integer polynomials of degree at most n with all coefficients at most B in modulus, and each has at most n roots. So the algebraic numbers are a countable union of finite sets, hence countable. Since $\mathbb{R}$ is uncountable, the complement is uncountable. $\square$

This argument is Cantor's, and it is beautiful but entirely non-constructive: it tells us that transcendentals abound while showing us none. Historically, the first proof came earlier, and is due to Liouville. It is longer, but it produces actual examples. The key point is that algebraic numbers are *badly approximable* by rationals.

Theorem 11.6 (Liouville)

Let α be an irrational root of a polynomial $P(X) = a_n X^n + \cdots + a_0$ with $a_j \in \mathbb{Z}$, $n \geq 1$ and $a_n \neq 0$. Then there is a constant $c = c(\alpha) > 0$ such that

$$\left|\alpha - \frac{p}{q}\right| \geq \frac{c}{q^n}$$

for all $p, q \in \mathbb{Z}$ with $q \geq 1$.

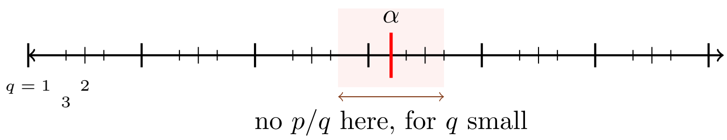
Proof. Since P has finitely many roots, we can find $\delta > 0$ such that P has no root other than α in $J = [\alpha - \delta, \alpha + \delta]$. As P' is continuous on the compact set J, it is bounded there, say $|P'| \leq M$ with $M > 0$; by the mean value theorem $|P(t) - P(s)| \leq M|t - s|$ for $s, t \in J$.

Let $p, q \in \mathbb{Z}$ with $q \geq 1$. Suppose first that $p/q \in J$. Then $P(p/q) \neq 0$, since α is the only root of P in J and α is irrational. But $q^n P(p/q) = \sum_j a_j p^j q^{n-j}$ is an integer, so $|q^n P(p/q)| \geq 1$ and hence $|P(p/q)| \geq q^{-n}$. Therefore

$$\left|\alpha - \frac{p}{q}\right| \geq \frac{1}{M}\,|P(\alpha) - P(p/q)| = \frac{|P(p/q)|}{M} \geq \frac{1}{Mq^n}.$$

If instead $p/q \notin J$ then $\left|\alpha - \frac{p}{q}\right| > \delta \geq \delta/q^n$. So $c = \min\{1/M, \delta\}$ works. $\square$

The content of the theorem is best seen geometrically. Rationals with denominator q are spaced $1/q$ apart, so we should expect to be able to find one within $1/(2q)$ of α, and the theory of continued fractions in the next section will do much better, giving one within $1/q^2$. Liouville's theorem says that for an algebraic α of degree n we can never do better than c/q^n: around each algebraic number there is a forbidden zone which the good rational approximations may not enter.



Turning this on its head: if a number *can* be approximated absurdly well by rationals, it cannot be algebraic.

Theorem 11.7

The number $\gamma = \displaystyle\sum_{j=0}^{\infty} \frac{1}{10^{j!}} = 1.110001000000000000000001\ldots$ is transcendental.

Proof. First, γ is irrational: its decimal expansion consists of 1's in the places $j!$ and 0's elsewhere, and so is neither terminating nor eventually periodic (the gaps between consecutive 1's grow without bound).

Let $N \geq 1$ and put $\frac{p_N}{q_N} = \sum_{j=0}^{N} 10^{-j!}$, where $q_N = 10^{N!}$ and $p_N \in \mathbb{N}$. Then

$$0 < \gamma - \frac{p_N}{q_N} = \sum_{j=N+1}^{\infty} 10^{-j!} \leq \sum_{k=(N+1)!}^{\infty} 10^{-k} = \frac{10}{9} \cdot 10^{-(N+1)!} \leq \frac{2}{q_N^{N+1}},$$

using $(N+1)! = (N+1) \cdot N!$, so that $10^{-(N+1)!} = q_N^{-(N+1)}$.

Suppose γ were algebraic, of degree n say, and let $c > 0$ be as in Theorem 11.6. Then for every N,

$$\frac{c}{q_N^n} \leq \left|\gamma - \frac{p_N}{q_N}\right| \leq \frac{2}{q_N^{N+1}}, \quad \text{so} \quad c \leq \frac{2}{q_N^{N+1-n}}.$$

Taking $N \geq n+1$ and letting $N \to \infty$, the right-hand side tends to 0 (as $q_N \to \infty$), contradicting $c > 0$. $\square$

Remark. Replacing γ by $\sum_{j\geq 0} \varepsilon_j 10^{-j!}$ with $\varepsilon_j \in \{1, 2\}$ gives an uncountable family of such numbers, all transcendental by the same argument. So Liouville's method also proves that there are uncountably many transcendentals, and this time we can point at them.

Numbers which admit approximations of this quality are called **Liouville numbers**. They are, in a precise sense, very rare – the set of them has Lebesgue measure zero – yet they are the only transcendentals we can construct with bare hands. As one might expect, proving that a *naturally occurring* number is transcendental is much harder. Hermite proved that e is transcendental, and Lindemann adapted his method to show that π is transcendental, thereby settling the two-thousand-year-old problem of squaring the circle in the negative. These proofs are not part of this course; Alan Baker's *Transcendental Number Theory* contains accessible (but miraculous) versions of both.

§12 The Baire Category Theorem

The theorem of this section is much more useful than its somewhat abstract formulation makes it appear. The informal version is this. We work in a non-empty complete metric space (X, d), and we have a list of properties $P_1, P_2, \ldots$ which points might have. Suppose each P_j is *stable* (if x has P_j then so does everything sufficiently close to x) while $\neg P_j$ is *unstable* (however close we look to any point, we can find something with P_j). Then there is a single point which has *all* of the properties at once.

That is a remarkable thing to be able to say. It means that to prove the existence of an object with infinitely many strange properties, it is enough to show that each property individually is achievable by a small perturbation. We shall use it to construct nowhere differentiable functions, and to construct compact sets which are simultaneously very thin and very rich.

Theorem 12.1 (Baire Category Theorem)
Let (X, d) be a non-empty complete metric space and let $U_1, U_2, \ldots$ be open sets whose complements have empty interior. Then $\bigcap_{j \geq 1} U_j \neq \emptyset$.

Note that 'U_j is open and U_j^c has empty interior' says exactly that U_j is open and dense.

Proof. Since U_1^c has empty interior, $U_1 \neq \emptyset$; choose $x_1 \in U_1$, and since U_1 is open choose $\delta_1 \in (0, 1)$ with $\overline{B}(x_1, 4\delta_1) \subseteq U_1$.

Now proceed inductively. Given x_n and $\delta_n > 0$, the ball $B(x_n, \delta_n)$ is not contained in U_{n+1}^c (which has empty interior), so we may choose $x_{n+1} \in U_{n+1}$ with $d(x_n, x_{n+1}) < \delta_n$. Since U_{n+1} is open we may then choose δ_{n+1} with $0 < \delta_{n+1} \leq \delta_n/4$ and $\overline{B}(x_{n+1}, 4\delta_{n+1}) \subseteq U_{n+1}$.

Then $d(x_n, x_{n+1}) < \delta_n \leq 4^{1-n}\delta_1$, so $\sum_n d(x_n, x_{n+1}) < \infty$ and (x_n) is Cauchy; by completeness $x_n \to x$ for some $x \in X$. Moreover, for $m > n$,

$$d(x_n, x_m) \leq \sum_{r=n}^{m-1} d(x_r, x_{r+1}) \leq \delta_n \left(1 + \tfrac{1}{4} + \tfrac{1}{16} + \cdots\right) \leq 2\delta_n,$$

so letting $m \to \infty$ gives $d(x_n, x) \leq 2\delta_n$, whence $x \in \overline{B}(x_n, 4\delta_n) \subseteq U_n$. As n was arbitrary, $x \in \bigcap_n U_n$. $\square$

Taking complements gives an equivalent statement which is often the more convenient one.

Theorem 12.2
Let (X, d) be a non-empty complete metric space and let $F_1, F_2, \ldots$ be closed sets with empty interior. Then $\bigcup_{j \geq 1} F_j \neq X$.

And the informal version we began with is the following.

Theorem 12.3
Let (X, d) be a non-empty complete metric space and let $P_1, P_2, \ldots$ be properties such that for each j:

(i) if x has property P_j then there is an $\varepsilon > 0$ such that every y with $d(x, y) < \varepsilon$ has property P_j;

(ii) given $x \in X$ and $\varepsilon > 0$, we can find y with $d(x, y) < \varepsilon$ having property P_j.

Then there is an $x_0 \in X$ having every property $P_1, P_2, \ldots$.

Proof. Let $U_j = \{x : x \text{ has } P_j\}$. Condition (i) says U_j is open and (ii) says U_j is dense, so U_j^c is closed with empty interior. Apply Theorem 12.1. $\square$

Baire's theorem has given rise to some standard, and rather unfortunate, terminology.

Definition 12.4 (Category)

Let (X, d) be a metric space.

(i) A set $E \subseteq X$ is **nowhere dense** if $\operatorname{Cl} E$ has empty interior.

(ii) A set E is of **first category** if it is contained in a countable union of closed sets with empty interior.

(iii) A set which is not of first category is said to be of **second category**.

The nomenclature is poor – there is nothing categorical about it, and 'second category' tells one nothing – so I shall try to avoid saying 'second category' and speak of 'not of first category' instead. If everything outside a set of first category has some property, I shall say that **quasi-all** points have that property. Sets of first category are the topological analogue of null sets: small, but small in a sense which has nothing to do with measure.

Lemma 12.5

Let (X, d) be a metric space.

(i) If X is non-empty and complete and $E \subseteq X$ is of first category, then $E \neq X$; indeed $E^c \neq \emptyset$.

(ii) A countable union of sets of first category is of first category.

Proof. (i) is Theorem 12.2. (ii) is because a countable union of countable unions is a countable union. □

§12.1 Uncountability, Again

Definition 12.6

Let (X, d) be a metric space and $E \subseteq X$. A point $x \in E$ is **isolated** in E if there is a $\delta > 0$ with $B(x, \delta) \cap E = \{x\}$. We say E has **no isolated points** if no point of E is isolated in E.

Theorem 12.7

A non-empty complete metric space with no isolated points is uncountable.

Proof. Suppose $X = \{x_1, x_2, \ldots\}$ is countable. Each singleton $F_n = \{x_n\}$ is closed, and has empty interior precisely because x_n is not isolated (every ball about x_n contains another point of X). By Theorem 12.2, $\bigcup_n F_n \neq X$, contradicting the enumeration. □

Corollary 12.8

$\mathbb{R}$ is uncountable. More generally, a non-empty closed subset of $\mathbb{R}$ with no isolated points is uncountable.

Proof. $\mathbb{R}$ is complete with no isolated points. A closed subset of a complete space is complete. □

This proof is much closer to Cantor's original argument than the diagonal argument of Part IA, and it has the advantage of avoiding extraneous notions like decimal expansions.

§12.2 Nowhere Differentiable Functions

Banach was a master of the Baire category theorem, and the following is one of his results. Recall from Theorem 6.6 that $C([0,1])$ with the uniform norm is a non-empty complete metric space.

Theorem 12.9 (Banach)

The set of functions in $C([0,1])$ which are differentiable at at least one point is of first category. In particular, continuous nowhere differentiable functions exist.

The key is to replace 'differentiable somewhere' by something closed.

Lemma 12.10

If $f : [0,1] \to \mathbb{R}$ is continuous and differentiable at some $x \in [0,1]$, then there is an $M > 0$ with $|f(x) - f(s)| \leq M|x - s|$ for all $s \in [0,1]$.

Proof. Since $\frac{f(s)-f(x)}{s-x} \to f'(x)$ as $s \to x$, there is a $\delta > 0$ such that $\left|\frac{f(s)-f(x)}{s-x} - f'(x)\right| \leq 1$ whenever $0 < |s - x| < \delta$, giving $|f(s) - f(x)| \leq (1 + |f'(x)|)|s - x|$ there. If $|s - x| \geq \delta$ then $|f(s) - f(x)| \leq 2\|f\|_\infty \leq \frac{2\|f\|_\infty}{\delta}|s - x|$. Take M to be the larger of the two constants. □

So let

$$E_m = \{f \in C([0,1]) : \exists x \in [0,1] \text{ with } |f(x) - f(s)| \leq m|x - s| \text{ for all } s \in [0,1]\}.$$

By Lemma 12.10, every function differentiable somewhere lies in some E_m. So it suffices to show each E_m is closed with empty interior.

Lemma 12.11

E_m is closed in $C([0,1])$.

Proof. Suppose $f_r \in E_m$ and $\|f_r - f\|_\infty \to 0$. For each r pick $x_r \in [0,1]$ with $|f_r(x_r) - f_r(s)| \leq m|x_r - s|$ for all s. By compactness of $[0,1]$, pass to a subsequence with $x_{r(k)} \to x$. Then for any $s \in [0,1]$, writing r for $r(k)$,

$$\begin{aligned}|f(x) - f(s)| &\leq |f(x) - f(x_r)| + |f(x_r) - f_r(x_r)| + |f_r(x_r) - f_r(s)| + |f_r(s) - f(s)| \\ &\leq |f(x) - f(x_r)| + 2\|f - f_r\|_\infty + m|x_r - s|.\end{aligned}$$

Letting $k \to \infty$, the first term tends to 0 by continuity of f, the second by uniform convergence, and the third to $m|x - s|$. Hence $|f(x) - f(s)| \leq m|x - s|$ for all s, so $f \in E_m$. □

Lemma 12.12

E_m has empty interior: given $f \in C([0,1])$ and $\varepsilon > 0$, there is an $h \notin E_m$ with $\|h - f\|_\infty < \varepsilon$.

Proof. The idea is to move first from f to a very nice function, and then from the nice function to a very bad one.

By the Weierstrass approximation theorem there is a polynomial g with $\|g - f\|_\infty < \varepsilon/3$. Being a polynomial, g is continuously differentiable, so $|g'| \leq M$ on $[0,1]$ for some M, and hence $|g(s) - g(t)| \leq M|s-t|$ for all s, t.

Now set

$$h(t) = g(t) + \frac{\varepsilon}{3}\cos(2\pi N t)$$

for an integer N to be chosen. Certainly $\|h - f\|_\infty < \varepsilon$. Let $t_k = k/(2N)$ for $0 \leq k \leq 2N$, so that $\cos(2\pi N t_k) = (-1)^k$ and

$$|h(t_{k+1}) - h(t_k)| = \left| g(t_{k+1}) - g(t_k) \mp \frac{2\varepsilon}{3} \right| \geq \frac{2\varepsilon}{3} - \frac{M}{2N}.$$

Choose N so large that $M/(2N) \leq \varepsilon/3$ and $N > 3m/\varepsilon$. Then $|h(t_{k+1}) - h(t_k)| \geq \varepsilon/3$ for every k.

Now let $x \in [0,1]$ be arbitrary, and pick k with $t_k \leq x \leq t_{k+1}$. By the triangle inequality

$$\max\left\{|h(t_k) - h(x)|,\ |h(t_{k+1}) - h(x)|\right\} \geq \frac{1}{2}|h(t_{k+1}) - h(t_k)| \geq \frac{\varepsilon}{6},$$

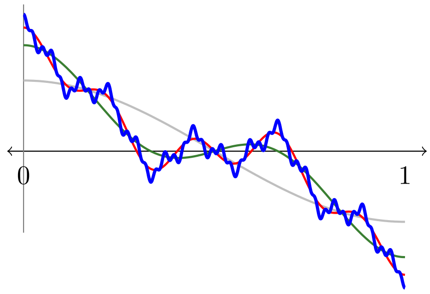
while $|t_k - x|, |t_{k+1} - x| \leq \frac{1}{2N} < \frac{\varepsilon}{6m}$. So for some $s \in \{t_k, t_{k+1}\}$ we have $|h(x) - h(s)| \geq \varepsilon/6 > m|x-s|$. Since x was arbitrary, $h \notin E_m$. $\square$

Proof of Theorem 12.9. By Lemmas 12.11 and 12.12 each E_m is closed with empty interior, so $\bigcup_m E_m$ is of first category and, by Lemma 12.5, is not all of $C([0,1])$. Any $f \notin \bigcup_m E_m$ is, by Lemma 12.10, differentiable nowhere. $\square$

Notice what a strange proof this is. We have shown that quasi-all continuous functions are nowhere differentiable – that the differentiable functions, which are the only ones anybody ever draws, form a vanishingly small part of $C([0,1])$ – without exhibiting a single nowhere differentiable function. One can of course write one down: Weierstrass's original example was

$$W(t) = \sum_{k=0}^{\infty} a^k \cos(b^k \pi t)$$

for suitable $0 < a < 1$ and odd integers b with $ab > 1$. Here are the partial sums for $a = 1/2$, $b = 3$ up to $k = 0, 1, 2$ and 4. Each new term adds wiggles which are smaller in amplitude but far steeper, and in the limit the graph has a corner at every point.



§12.3 Perfect Nowhere Dense Sets

Here is a second, and rather different, application. A closed subset of $\mathbb{R}$ with no isolated points is uncountable, by the corollary above. Are there such sets which are nevertheless nowhere dense – containing no interval at all? The Cantor set is one; but we can prove existence without constructing anything, by applying Baire's theorem in a space of *sets*.

Let $\mathcal{K}$ denote the collection of non-empty compact subsets of $[0,1]$, with the Hausdorff metric ρ of the previous section. Since the compact subsets of $[0,1]$ form a closed subset of the space of compact subsets of $\mathbb{R}$, Theorem 9.8 tells us that $(\mathcal{K},\rho)$ is a non-empty complete metric space.

Lemma 12.13

For $k \geq 1$ let $\mathcal{E}_k = \{E \in \mathcal{K} : \exists\, x \in E \text{ with } (x - \frac{1}{k}, x + \frac{1}{k}) \cap E = \{x\}\}$.

(i) $\mathcal{E}_k$ is closed in $(\mathcal{K},\rho)$.

(ii) $\mathcal{E}_k$ has empty interior.

(iii) The collection of compact sets possessing an isolated point is of first category in $\mathcal{K}$.

Proof. (i) Let $E_r \in \mathcal{E}_k$ with $\rho(E_r, E) \to 0$, and pick $x_r \in E_r$ with $(x_r - \frac{1}{k}, x_r + \frac{1}{k}) \cap E_r = \{x_r\}$. Passing to a subsequence, $x_r \to x$; since $d(x_r, E) \leq \sigma(E_r, E) \to 0$ and E is closed, $x \in E$.

Suppose $y \in E$ with $y \neq x$ and $|x - y| < \frac{1}{k}$. Put $4\eta = \frac{1}{k} - |x-y| > 0$. Since $\sigma(E, E_r) \to 0$ we can find $y_r \in E_r$ with $|y_r - y| \to 0$. For r large we have $|x_r - y_r| < |x-y| + 2\eta < \frac{1}{k}$ and $y_r \neq x_r$ (as $|x-y| > 0$), so y_r lies in $(x_r - \frac{1}{k}, x_r + \frac{1}{k}) \cap E_r$ and differs from x_r. This contradiction shows $(x - \frac{1}{k}, x + \frac{1}{k}) \cap E = \{x\}$, so $E \in \mathcal{E}_k$.

(ii) Let $E \in \mathcal{K}$ and $\varepsilon > 0$. Take $0 < \eta < \varepsilon$ and set $F = \{x \in [0,1] : d(x, E) \leq \eta\}$, which is a non-empty compact subset of $[0,1]$ with $E \subseteq F$, so $\sigma(E, F) = 0$ and $\sigma(F, E) \leq \eta$, giving $\rho(E, F) \leq \eta < \varepsilon$. But F has no isolated points at all: if $x \in F$, take $e \in E$ with $|x - e| \leq \eta$; the whole segment between e and x lies in F, and if $x = e$ then one of $[x, x+\eta] \cap [0,1]$, $[x - \eta, x] \cap [0,1]$ is a non-degenerate interval contained in F. Either way every neighbourhood of x meets $F \setminus \{x\}$. So $F \notin \mathcal{E}_k$.

(iii) A compact set has an isolated point if and only if it lies in $\mathcal{E}_k$ for some k, so the collection in question is $\bigcup_k \mathcal{E}_k$, a countable union of closed sets with empty interior. $\square$

Lemma 12.14

For integers $k \geq 1$ and $0 \leq j < k$, let $\mathcal{F}_{j,k} = \{E \in \mathcal{K} : E \supseteq [\frac{j}{k}, \frac{j+1}{k}]\}$. Then $\mathcal{F}_{j,k}$ is closed with empty interior, and the collection of compact subsets of $[0,1]$ with non-empty interior is of first category.

Proof. Write $I = [\frac{j}{k}, \frac{j+1}{k}]$. If $E_r \in \mathcal{F}_{j,k}$ and $\rho(E_r, E) \to 0$, then for $y \in I$ we have $y \in E_r$, so $d(y, E) \leq \sigma(E_r, E) \to 0$ and hence $y \in E$; thus $E \in \mathcal{F}_{j,k}$ and $\mathcal{F}_{j,k}$ is closed.

For empty interior, let $E \in \mathcal{K}$ and $\varepsilon > 0$. If $E \notin \mathcal{F}_{j,k}$ we are done. Otherwise $E \supseteq I$; let a be the midpoint of I, choose $0 < \eta < \min\{\varepsilon, \frac{1}{2k}\}$ and set $F = E \setminus (a-\eta, a+\eta)$. Then F is compact and non-empty (it still contains the endpoints of I), $\sigma(F, E) = 0$, and every removed point is within η of $a \pm \eta \in F$, so $\sigma(E, F) \leq \eta$ and $\rho(E, F) \leq \eta < \varepsilon$. But $a \notin F$, so $F \notin \mathcal{F}_{j,k}$.

Finally, a compact $E \subseteq [0,1]$ has non-empty interior if and only if it contains some non-degenerate interval, hence some $[\frac{j}{k}, \frac{j+1}{k}]$; so the collection of such E is $\bigcup_{j,k} \mathcal{F}_{j,k}$. □

Theorem 12.15

Quasi-all non-empty compact subsets of $[0,1]$ have empty interior and no isolated points. In particular such sets exist, and by Theorem 12.7 they are uncountable.

Proof. By Lemmas 12.13 and 12.14 the sets we wish to exclude form a union of two sets of first category, hence a set of first category, which by Lemma 12.5 is not all of $\mathcal{K}$. □

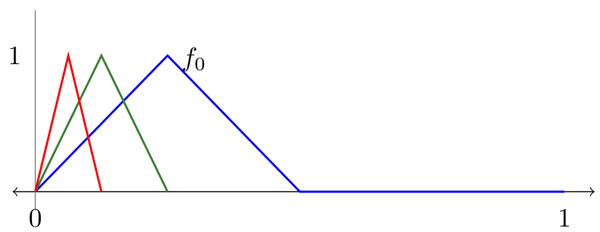
Sets which are closed with no isolated points are called *perfect*; if they make pretty pictures they are called fractals. The reader is not obliged to remember either name.

§12.4 Pointwise Convergence and Boundedness

Our last application concerns the gap between pointwise and uniform convergence. Recall the 'witch's hat' functions $f_n : [0,1] \to \mathbb{R}$,

$$f_n(x) = \begin{cases} 2^{n+1}x & 0 \leq x \leq 2^{-(n+1)}, \\ 2 - 2^{n+1}x & 2^{-(n+1)} < x \leq 2^{-n}, \\ 0 & \text{otherwise}, \end{cases}$$

which satisfy $f_n(x) \to 0$ for every x while $\|f_n\|_\infty = 1$ for every n: the hat becomes narrower but never shorter, and slides off to the left.



The trouble is confined to a shrinking interval, and one might hope that pointwise convergence forces uniform convergence somewhere. Alas, it does not.

Lemma 12.16

There exist $g_n \in C([0,1])$ with $\|g_n\|_\infty \le 1$ and $g_n(x) \to 0$ for every $x \in [0,1]$, such that (g_n) fails to converge uniformly on every non-degenerate interval $I \subseteq [0,1]$.

Proof. Enumerate the subintervals of $[0,1]$ with distinct rational endpoints as $[a_1, b_1], [a_2, b_2], \ldots$, and let $p_1, p_2, \ldots$ be distinct primes. Define $g_m = 0$ unless $m = p_k^r$ for some $k, r \ge 1$ (which determines k and r uniquely), and in that case

$$g_{p_k^r}(x) = \begin{cases} 2^{-k} f_r\left(\dfrac{x - a_k}{b_k - a_k}\right) & x \in [a_k, b_k], \\ 0 & \text{otherwise.} \end{cases}$$

Since $f_r(0) = f_r(1) = 0$, each g_m is continuous, and clearly $\|g_m\|_\infty \le 2^{-k} \le 1$.

Pointwise convergence: fix x and $\varepsilon > 0$, and choose K with $2^{-K} < \varepsilon$. For $k > K$ we have $|g_{p_k^r}(x)| \le 2^{-k} < \varepsilon$ for every r. For each of the finitely many $k \le K$, we have $f_r(y) \to 0$ as $r \to \infty$ for each fixed y, so there is an R_k with $|g_{p_k^r}(x)| < \varepsilon$ for $r \ge R_k$. Hence $|g_m(x)| \ge \varepsilon$ for at most finitely many m, and $g_m(x) \to 0$.

Non-uniformity: any non-degenerate interval I contains some $[a_k, b_k]$, and $\sup_{x \in [a_k, b_k]} |g_{p_k^r}(x)| = 2^{-k}$ for every r, so g_m does not tend to 0 uniformly on I. $\square$

In spite of this, Baire's theorem gives a remarkable positive result: the failure can never be one of *size*.

Theorem 12.17 (Osgood)

Let $f_n \in C([0,1])$ with $f_n(x) \to 0$ for every $x \in [0,1]$. Then given $\varepsilon > 0$ there is a non-degenerate interval $(a,b) \subseteq [0,1]$ and an N such that $|f_n(x)| \le \varepsilon$ for all $n \ge N$ and all $x \in (a,b)$.

Proof. For $n \ge 1$ set

$$E_n = \{x \in [0,1] : |f_m(x)| \le \varepsilon \text{ for all } m \ge n\} = \bigcap_{m \ge n} f_m^{-1}([-\varepsilon, \varepsilon]),$$

an intersection of closed sets and hence closed. Since $f_m(x) \to 0$ for each x, every x lies in some E_n, so $\bigcup_n E_n = [0,1]$. Now $[0,1]$ is a non-empty complete metric space, so by Theorem 12.2 not every E_n can have empty interior. Take N with E_N containing an interval (a,b). $\square$

Corollary 12.18 (Uniform Boundedness)

Let $f_n \in C([0,1])$ with $f_n(x) \to 0$ for every x. Then there is a non-degenerate interval (a,b) and an $M > 0$ with $|f_n(x)| \le M$ for all $n \ge 1$ and all $x \in (a,b)$.

> *Proof.* Apply Theorem 12.17 with $\varepsilon = 1$ to get (a,b) and N with $|f_n| \le 1$ on (a,b) for $n \ge N$. The finitely many remaining functions $f_1, \dots, f_{N-1}$ are each bounded on $[0,1]$, so $M = \max\{1, \|f_1\|_\infty, \dots, \|f_{N-1}\|_\infty\}$ works. $\square$

Comparing this with Lemma 12.16, we see that the functions g_n constructed there are as bad as pointwise-convergent sequences are permitted to be: they refuse to converge uniformly on any interval, but they cannot be allowed to blow up on every interval as well. This is the elementary ancestor of the principle of uniform boundedness in Part II Linear Analysis, which is proved by exactly the same Baire category argument in a Banach space.

§13 Continued Fractions

We are used to writing real numbers as decimals, but there are other ways of specifying a real number which are often more convenient and always more interesting. The oldest is the method of continued fractions.

Suppose x is irrational with $0 < x < 1$. Then $1/x > 1$ is irrational, so there is a unique integer $N(x) \ge 1$ with

$$\frac{1}{N(x)+1} < x < \frac{1}{N(x)}, \qquad \text{that is} \qquad N(x) = \left\lfloor \frac{1}{x} \right\rfloor,$$

and setting $T(x) = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor$ we have $0 < T(x) < 1$, $T(x)$ irrational, and

$$x = \frac{1}{N(x) + T(x)}.$$

We may now do the same thing to $T(x)$, obtaining $T(x) = \frac{1}{NT(x)+T^2(x)}$, and so

$$x = \cfrac{1}{N(x) + \cfrac{1}{NT(x) + T^2(x)}} = \cfrac{1}{N(x) + \cfrac{1}{NT(x) + \cfrac{1}{NT^2(x) + T^3(x)}}} = \cdots,$$

and so on indefinitely. We call

$$\cfrac{1}{N(x) + \cfrac{1}{NT(x) + \cfrac{1}{NT^2(x) + \cfrac{1}{NT^3(x) + \ddots}}}}$$

the **continued fraction expansion** of x. For the moment this is nothing more than a pretty way of writing down the sequence of positive integers $N(x), NT(x), NT^2(x), \dots$; in the next subsection we shall show that it can be assigned a numerical value, and that the value is x. For a general irrational y we prefix the integer part, writing the expansion of y as $\lfloor y \rfloor$ followed by the expansion of $y - \lfloor y \rfloor$.

The same procedure works for rationals, except that it may stop. Let us see why.

Example 13.1

Take $x = 100/37$. Then

$$\frac{100}{37} = 2 + \frac{26}{37} = 2 + \frac{1}{\frac{37}{26}} = 2 + \frac{1}{1 + \frac{11}{26}} = 2 + \frac{1}{1 + \frac{1}{\frac{26}{11}}} = 2 + \frac{1}{1 + \frac{1}{2 + \frac{4}{11}}}$$

$$= 2 + \frac{1}{1 + \frac{1}{2 + \frac{1}{2 + \frac{3}{4}}}} = 2 + \frac{1}{1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{1 + \frac{1}{3}}}}} = 2 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{1}{3}}}}}.$$

Lemma 13.2

The continued fraction algorithm applied to a rational number is Euclid's algorithm, and in particular it terminates.

Proof. Suppose at some stage we are looking at r_k/s_k with $0 < r_k < s_k$ coprime positive integers, and we write

$$\frac{r_k}{s_k} = \frac{1}{a_k + \frac{r_{k+1}}{s_{k+1}}}, \qquad a_k = \left\lfloor \frac{s_k}{r_k} \right\rfloor.$$

Then $s_k/r_k = a_k + r_{k+1}/s_{k+1}$, so $s_{k+1} = r_k$ and $s_k = a_k s_{k+1} + r_{k+1}$ with $0 \leq r_{k+1} < s_{k+1}$. That is exactly one step of the Euclidean algorithm applied to the pair (s_k, r_k), and r_{k+1} is the remainder. Since $s_1 > s_2 > \cdots$ is a strictly decreasing sequence of positive integers, the process stops. (Coprimality is preserved, again as in Euclid's algorithm.) $\square$

Consequently a number with a non-terminating continued fraction expansion is irrational, and this gives a rather satisfying proof of an old fact.

Theorem 13.3

$\sqrt{2}$ is irrational.

Proof. We have $\sqrt{2} = 1 + (\sqrt{2} - 1)$ and

$$\sqrt{2} - 1 = \frac{(\sqrt{2} - 1)(\sqrt{2} + 1)}{\sqrt{2} + 1} = \frac{1}{1 + \sqrt{2}} = \frac{1}{2 + (\sqrt{2} - 1)}.$$

So writing $\beta = \sqrt{2} - 1 \in (0, 1)$ we have $\beta = \frac{1}{2+\beta}$, and the continued fraction algorithm applied to β produces $N(\beta) = 2$ and $T(\beta) = \beta$ again. It therefore never terminates, giving

$$\sqrt{2} = 1 + \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{2 + \ddots}}},$$

and by Lemma 13.2 the number $\sqrt{2}$ cannot be rational. $\square$

§13.1 Gauss's Observation

Before we make sense of the value of a continued fraction, here is a beautiful aside. The decimal expansion has a map behind it too: if $x \in [0,1)$ and $Dx = 10x - \lfloor 10x \rfloor$, then the digits of x are read off by iterating D, and if X is uniformly distributed on $[0,1]$ then so is DX – which is why the digits of a 'random' number are independent and uniform.

What is the analogue for continued fractions? The map T does *not* preserve the uniform distribution. Gauss found the distribution it does preserve.

Lemma 13.4 (Gauss)
Let X be a random variable on $[0,1)$ with density

$$f(x) = \frac{1}{\log 2} \cdot \frac{1}{1+x}.$$

Then TX has the same density.

Proof. First, f is a density: $\int_0^1 \frac{\mathrm{d}x}{\log 2\,(1+x)} = \frac{\log 2}{\log 2} = 1$. Now for $0 \le a \le 1$,

$$\Pr(TX \le a) = \Pr\left(\frac{1}{X} - \left\lfloor \frac{1}{X} \right\rfloor \le a\right) = \sum_{n=1}^{\infty} \Pr\left(n \le \frac{1}{X} \le n+a\right) = \sum_{n=1}^{\infty} \Pr\left(\frac{1}{n+a} \le X \le \frac{1}{n}\right)$$
$$= \frac{1}{\log 2} \sum_{n=1}^{\infty} \int_{1/(n+a)}^{1/n} \frac{\mathrm{d}x}{1+x} = \frac{1}{\log 2} \sum_{n=1}^{\infty} \left[\log\left(1 + \frac{1}{n}\right) - \log\left(1 + \frac{1}{n+a}\right)\right].$$

Both parts telescope. Summing to N,

$$\sum_{n=1}^{N} \big(\log(n+1) - \log n\big) = \log(N+1), \qquad \sum_{n=1}^{N} \big(\log(n+1+a) - \log(n+a)\big) = \log(N+1+a) - \log(1+a),$$

so the partial sum is $\log(N+1) - \log(N+1+a) + \log(1+a) \to \log(1+a)$ as $N \to \infty$.
Hence

$$\Pr(TX \le a) = \frac{\log(1+a)}{\log 2} = \int_0^a f(x)\,\mathrm{d}x.$$ $\square$

Corollary 13.5

With X as above, for each $m \geq 0$ and each $j \geq 1$,

$$\Pr(NT^m X = j) = \frac{1}{\log 2}\int_{1/(j+1)}^{1/j} \frac{\mathrm{d}x}{1+x} = \frac{1}{\log 2}\log\left(\frac{(j+1)^2}{j(j+2)}\right) \sim \frac{1}{j^2 \log 2}$$

as $j \to \infty$.

Proof. By Lemma 13.4, $T^m X$ has the same density as X, and $NY = j$ if and only if $\frac{1}{j+1} < Y \leq \frac{1}{j}$. Evaluating the integral gives $\frac{1}{\log 2}\left(\log\frac{j+1}{j} - \log\frac{j+2}{j+1}\right)$, which is the stated logarithm; and $\log\frac{(j+1)^2}{j(j+2)} = \log\left(1 + \frac{1}{j^2+2j}\right) \sim j^{-2}$. $\square$

With a little more work (which we shall not do) one can show that if X has Gauss's density then $NX, NTX, NT^2X, \ldots$ are independent and identically distributed; and it is true, though harder, that if Y is merely *uniform* on $[0,1]$ then the proportion of the first M partial quotients of Y which equal j still tends to the above value. So in a 'typical' continued fraction the digit 1 appears about 41% of the time, and large partial quotients are rare.

§13.2 Convergents

We now return to the question of what a continued fraction *means*.

Definition 13.6

Let a_0 be an integer and $a_1, a_2, \ldots$ strictly positive integers. The associated **convergents** are the rationals

$$\frac{p_n}{q_n} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}}}.$$

Evaluating from the bottom up is the natural thing to try. Write $\frac{r_k}{s_k} = a_k + \cfrac{1}{a_{k+1} + \cfrac{1}{\ddots + \frac{1}{a_n}}}$,

so that $r_n = a_n$, $s_n = 1$, and

$$\frac{r_k}{s_k} = a_k + \frac{1}{r_{k+1}/s_{k+1}} = \frac{a_k r_{k+1} + s_{k+1}}{r_{k+1}}, \qquad \text{i.e.} \qquad \begin{pmatrix} r_k \\ s_k \end{pmatrix} = \begin{pmatrix} a_k & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} r_{k+1} \\ s_{k+1} \end{pmatrix}.$$

Hence

$$\begin{pmatrix} p_n \\ q_n \end{pmatrix} = \begin{pmatrix} r_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix}\cdots\begin{pmatrix} a_{n-1} & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} a_n \\ 1 \end{pmatrix}.$$

Similarly $\begin{pmatrix} p_{n-1} \\ q_{n-1} \end{pmatrix}$ is the same product with the last factor replaced by $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, so putting the two columns side by side,

$$\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix}\cdots\begin{pmatrix} a_n & 1 \\ 1 & 0 \end{pmatrix}.$$

This is the pay-off for recasting things in matrix form: every fact we want now falls out.

Theorem 13.7

With p_n, q_n as above (and the conventions $p_{-1} = 1$, $q_{-1} = 0$):

(i) $p_{n+1} = a_{n+1}p_n + p_{n-1}$ and $q_{n+1} = a_{n+1}q_n + q_{n-1}$;

(ii) $p_n q_{n-1} - q_n p_{n-1} = (-1)^{n+1}$, and in particular p_n and q_n are coprime;

(iii) $\dfrac{p_n}{q_n} - \dfrac{p_{n-1}}{q_{n-1}} = \dfrac{(-1)^{n+1}}{q_n q_{n-1}}$;

(iv) $q_n \geq F_{n+1} \to \infty$, where F_m is the mth Fibonacci number;

(v) $\dfrac{p_0}{q_0} < \dfrac{p_2}{q_2} < \dfrac{p_4}{q_4} < \cdots < \cdots < \dfrac{p_5}{q_5} < \dfrac{p_3}{q_3} < \dfrac{p_1}{q_1}$, and $p_n/q_n \to \alpha$ for some real α with $\left|\alpha - \frac{p_n}{q_n}\right| \leq \frac{1}{q_n q_{n+1}}$.

Proof. (i) Multiply the matrix identity for $n+1$ as

$$\begin{pmatrix} p_{n+1} & p_n \\ q_{n+1} & q_n \end{pmatrix} = \begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} \begin{pmatrix} a_{n+1} & 1 \\ 1 & 0 \end{pmatrix}$$

and read off the first column.

(ii) Take determinants: each factor $\left(\begin{smallmatrix} a_k & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has determinant -1, and there are $n+1$ of them, so $p_n q_{n-1} - q_n p_{n-1} = (-1)^{n+1}$. Any common factor of p_n and q_n would divide this.

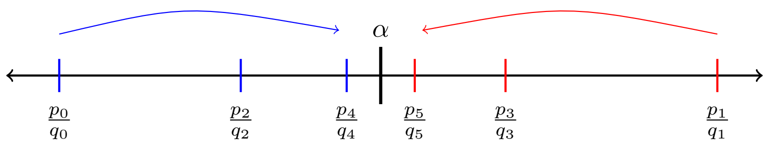
(iii) Divide (ii) by $q_n q_{n-1}$.

(iv) By (i), $q_{n+1} = a_{n+1}q_n + q_{n-1} \geq q_n + q_{n-1}$, with $q_0 = 1$ and $q_1 = a_1 \geq 1$, so induction gives $q_n \geq F_{n+1}$.

(v) From (i) and (iii),

$$\frac{p_{n+1}}{q_{n+1}} - \frac{p_{n-1}}{q_{n-1}} = \left(\frac{p_{n+1}}{q_{n+1}} - \frac{p_n}{q_n}\right) + \left(\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}}\right) = \frac{(-1)^{n+2}}{q_{n+1}q_n} + \frac{(-1)^{n+1}}{q_n q_{n-1}} = \frac{(-1)^{n+1}}{q_n}\left(\frac{1}{q_{n-1}} - \frac{1}{q_{n+1}}\right),$$

which has the sign of $(-1)^{n+1}$ since $q_{n+1} > q_{n-1}$. So the even convergents increase and the odd ones decrease. By (iii) every even convergent is below every odd one, so both are bounded monotone sequences and converge; and since $\left|\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}}\right| = \frac{1}{q_n q_{n-1}} \to 0$ by (iv), the two limits agree, say to α. As α lies between p_n/q_n and p_{n+1}/q_{n+1}, we get $\left|\alpha - \frac{p_n}{q_n}\right| \leq \frac{1}{q_n q_{n+1}}$. □



So a continued fraction does converge, and it converges in the pleasantest possible way: the convergents alternate around the limit, so each one gives an error bound for the previous one. It remains to check that when we start with a number x we get back to x.

Lemma 13.8

If $x \in (0,1)$ is irrational and $a_1 = N(x)$, $a_2 = NT(x)$, $\ldots$ are the partial quotients produced by the continued fraction algorithm (with $a_0 = 0$), then $p_n/q_n \to x$.

Proof. Truncating the algorithm at stage n gives x exactly as the continued fraction with entries $a_0, a_1, \ldots, a_n + T^n(x)$ in place of $a_0, \ldots, a_n$, where $0 < T^n(x) < 1$. Increasing a partial quotient a_n increases p_n/q_n when n is even and decreases it when n is odd (this is immediate from the fact, used repeatedly, that $t \mapsto \frac{1}{a+t}$ is decreasing on $t \geq 0$, and that these substitutions are nested). It follows that

$$\frac{p_{2k}}{q_{2k}} < x < \frac{p_{2k-1}}{q_{2k-1}}$$

for every k. As both sides tend to the limit α of Theorem 13.7(v), we get $x = \alpha$. $\square$

§13.3 Convergents Are the Best Approximations

The convergents are not merely good approximations: they are the best possible ones for their size of denominator.

Lemma 13.9

Let $u, v \in \mathbb{Z}$ with $1 \leq v \leq q_n$, and suppose $\left|\frac{u}{v} - \frac{p_{n+1}}{q_{n+1}}\right| \leq \frac{1}{q_n q_{n+1}}$. Then $v = q_n$ and (provided $q_{n+1} > 2$) $u = p_n$.

Proof. First, $\frac{u}{v} \neq \frac{p_{n+1}}{q_{n+1}}$: since p_{n+1}, q_{n+1} are coprime this would force $q_{n+1} \mid v$, impossible as $1 \leq v \leq q_n < q_{n+1}$. Hence $|uq_{n+1} - vp_{n+1}| \geq 1$ and

$$\left|\frac{u}{v} - \frac{p_{n+1}}{q_{n+1}}\right| = \frac{|uq_{n+1} - vp_{n+1}|}{vq_{n+1}} \geq \frac{1}{vq_{n+1}} \geq \frac{1}{q_n q_{n+1}}.$$

So both inequalities are equalities: $v = q_n$ and $|uq_{n+1} - q_n p_{n+1}| = 1$. But also $|p_n q_{n+1} - q_n p_{n+1}| = 1$ by Theorem 13.7(ii), so subtracting gives $(u - p_n)q_{n+1} \in \{0, \pm 2\}$, and if $q_{n+1} > 2$ this forces $u = p_n$. $\square$

Theorem 13.10

Let α be the value of the continued fraction with convergents p_n/q_n, and suppose $q_{n+1} > 2$. Then

$$\left|\alpha - \frac{p_n}{q_n}\right| \leq \left|\alpha - \frac{p}{q}\right|$$

for all integers p, q with $1 \leq q \leq q_n$.

Proof. Since α lies between $\frac{p_n}{q_n}$ and $\frac{p_{n+1}}{q_{n+1}}$, we have

$$\left|\alpha - \frac{p_n}{q_n}\right| + \left|\alpha - \frac{p_{n+1}}{q_{n+1}}\right| = \left|\frac{p_n}{q_n} - \frac{p_{n+1}}{q_{n+1}}\right| = \frac{1}{q_n q_{n+1}}.$$

Suppose $1 \le q \le q_n$ and $\left|\alpha - \frac{p}{q}\right| < \left|\alpha - \frac{p_n}{q_n}\right|$. Then

$$\left|\frac{p}{q} - \frac{p_{n+1}}{q_{n+1}}\right| \le \left|\frac{p}{q} - \alpha\right| + \left|\alpha - \frac{p_{n+1}}{q_{n+1}}\right| < \left|\alpha - \frac{p_n}{q_n}\right| + \left|\alpha - \frac{p_{n+1}}{q_{n+1}}\right| = \frac{1}{q_n q_{n+1}},$$

so by Lemma 13.9 we get $q = q_n$ and $p = p_n$, contradicting the strict inequality. □

Corollary 13.11

If x is irrational, there are infinitely many pairs of integers (u, v) with $v \ge 1$ and

$$\left|x - \frac{u}{v}\right| < \frac{1}{v^2}.$$

Proof. Take $u = p_n$, $v = q_n$. By Theorem 13.7(v) and $q_{n+1} > q_n$,

$$\left|x - \frac{p_n}{q_n}\right| \le \frac{1}{q_n q_{n+1}} < \frac{1}{q_n^2},$$

and $q_n \to \infty$ gives infinitely many distinct pairs. □

This should be compared with Liouville's Theorem 11.6, which says that an algebraic irrational of degree n satisfies $|x - p/q| \ge cq^{-n}$. For $n = 2$ the two statements are in direct competition, and neither can be improved to the point of contradicting the other: quadratic irrationals are approximable to order q^{-2} and no better.

§13.4 Badly Approximable Numbers

Large partial quotients a_j produce large jumps in $q_{n+1} = a_{n+1}q_n + q_{n-1}$, and hence unusually good approximations. So the most *resistant* number to rational approximation should be the one whose partial quotients are all as small as possible.

Example 13.12 (The Golden Ratio)

Let

$$\sigma = \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ddots}}}.$$

Then $\sigma = \frac{1}{1+\sigma}$, so $\sigma^2 + \sigma - 1 = 0$ and (taking the positive root) $\sigma = \frac{\sqrt{5}-1}{2} = 1/\varphi$, where $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.

Here every $a_j = 1$, so by Theorem 13.7(i) we have $p_{n+1} = p_n + p_{n-1}$ and $q_{n+1} = q_n + q_{n-1}$. With $a_0 = 0$ we get $p_0 = 0$, $q_0 = 1$, $p_1 = q_1 = 1$, and induction gives

$$p_n = F_n, \qquad q_n = F_{n+1},$$

where $F_0 = 0$, $F_1 = 1$, $F_{m+1} = F_m + F_{m-1}$ are the Fibonacci numbers. Theorem 13.7(ii) becomes the identity $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$.

Solving the recurrence, $F_m = \frac{1}{\sqrt{5}}(\varphi^m - (-\varphi)^{-m})$. Since $\sigma\varphi = 1$, a short computation gives

$$\sigma - \frac{F_n}{F_{n+1}} = \frac{\sigma F_{n+1} - F_n}{F_{n+1}} = \frac{(-1)^n \varphi^{-n}(1+\varphi^{-2})}{\sqrt{5}\, F_{n+1}},$$

and as $F_{n+1} \sim \varphi^{n+1}/\sqrt{5}$ this gives

$$\left|\sigma - \frac{p_n}{q_n}\right| \sim \frac{1}{\sqrt{5}\, q_n^2}.$$

So the convergents to σ approximate it to order q^{-2} and to no better order – the constant $1/\sqrt{5}$ is exactly what the general theory guarantees, and not a whit more. This lets us prove that σ is *badly approximable.*

Proposition 13.13

There is an $m > 0$ such that $\left|\sigma - \frac{p}{q}\right| > \frac{m}{q^2}$ for all integers p and all $q \geq 1$.

Proof. Fix p, q with $q \geq 1$, and let n be least with $q \leq q_n$ and $q_{n+1} > 2$. By Theorem 13.10,

$$\left|\sigma - \frac{p}{q}\right| \geq \left|\sigma - \frac{p_n}{q_n}\right| \geq \frac{c}{q_n^2}$$

for some constant $c > 0$, by Example 13.12. By minimality of n (and $q_n = q_{n-1} + q_{n-2} \leq 2q_{n-1}$) we have $q_n \leq 2q_{n-1} \leq 2q$ for all n beyond the first couple, so $\left|\sigma - \frac{p}{q}\right| \geq \frac{c}{4q^2}$. Adjusting the constant to deal with the finitely many small cases gives the result. □

Hardy and Wright's *An Introduction to the Theory of Numbers* contains a chapter on approximation by rationals in which, among many other things, it is shown that σ really is the most resistant of all irrationals in this sense. If I were asked to nominate a single book to be taken away by somebody leaving professional mathematics but wishing to keep up an interest in the subject, that book would be my first choice.

A Nice, But Starred, Formula

This subsection is not examinable, but it is too pretty to leave out. Continued fractions generalise in many directions; here is one of them. We are used to approximating a function by polynomials; sometimes it is better to approximate by a *rational* function P/Q, and if approximation by polynomials leads to Taylor series, approximation by rational functions leads to something like continued fractions. Lambert found the following.

Theorem 13.14 (Lambert)

Let

$$R_n(x) = \cfrac{x}{1 - \cfrac{x^2}{3 - \cfrac{x^2}{5 - \cfrac{x^2}{\ddots - \cfrac{x^2}{2n-1}}}}}.$$

Then $R_n(x) \to \tan x$ as $n \to \infty$ for all real x with $|x| \leq 1$.

The proof runs along lines which will now be familiar. One first generalises the matrix identity: if p_n, q_n are defined by

$$\begin{pmatrix} p_n & b_n p_{n-1} \\ q_n & b_n q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & b_0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & b_1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_n & b_n \\ 1 & 0 \end{pmatrix},$$

then p_n/q_n is the generalised continued fraction with numerators b_j and denominators a_j, and

$$p_n = a_n p_{n-1} + b_{n-1} p_{n-2}, \qquad q_n = a_n q_{n-1} + b_{n-1} q_{n-2}.$$

One then writes

$$S_n(x) = \frac{1}{2^n n!} \int_0^x (x^2 - t^2)^n \cos t \, \mathrm{d}t$$

and checks, by integrating by parts twice in the manner of Lemma 11.2, that $S_n(x) = q_n(x)\sin x - p_n(x)\cos x$ where p_n, q_n satisfy exactly the recurrences

$$p_n(x) = (2n-1)p_{n-1}(x) - x^2 p_{n-2}(x), \qquad q_n(x) = (2n-1)q_{n-1}(x) - x^2 q_{n-2}(x)$$

with $p_0 = 0$, $q_0 = 1$, $p_1(x) = x$, $q_1 = 1$. These are the recurrences for the continued fraction above, and the estimate $|S_n(x)| \leq \frac{|x|^{2n+1}}{2^n n!} \to 0$ then gives $p_n(x)/q_n(x) \to \tan x$.

§14 Winding Numbers

We all know that complex analysis has a great deal to say about the number of times a curve goes round a point. In this final section we make the notion precise without using any complex analysis at all, and then use it to give a second proof of two theorems from earlier in the course. The moral will be that parts of complex analysis are really topology in disguise.

§14.1 Lifting

Write $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$. The basic fact is that a continuous path on the circle can be followed continuously by an angle.

Theorem 14.1 (Path Lifting)

Let $g : [0,1] \to \mathbb{T}$ be continuous with $g(0) = e^{i\theta_0}$. Then there is a unique continuous $\theta : [0,1] \to \mathbb{R}$ with $\theta(0) = \theta_0$ and $g(t) = e^{i\theta(t)}$ for all t.

Proof. For *existence*, note first that on the half-plane $H = \{z : \operatorname{Re} z > 0\}$ we have a continuous argument function

$$\arg(u + iv) = \arctan(v/u) \in (-\tfrac{\pi}{2}, \tfrac{\pi}{2}),$$

satisfying $w = |w|e^{i \arg w}$ and $\arg 1 = 0$.

Now g is uniformly continuous by Theorem 1.12, so we may choose N with $|g(s) - g(t)| < 1$ whenever $|s - t| \leq 1/N$. Fix r and $t \in [\frac{r}{N}, \frac{r+1}{N}]$. Then $\left|\frac{g(t)}{g(r/N)} - 1\right| = |g(t) - g(r/N)| < 1$, and a point of $\mathbb{T}$ within distance 1 of 1 has positive real part, so $g(t)/g(r/N) \in H$. We may therefore define θ successively on the intervals $[\frac{r}{N}, \frac{r+1}{N}]$ by

$$\theta(t) = \theta\big(\tfrac{r}{N}\big) + \arg\left(\frac{g(t)}{g(r/N)}\right),$$

starting from $\theta(0) = \theta_0$. This is continuous on each interval, agrees at the shared endpoints, and satisfies $e^{i\theta(t)} = e^{i\theta(r/N)} \cdot \frac{g(t)}{g(r/N)} = g(t)$ by induction on r.

For *uniqueness*, suppose $\psi, \phi : [0,1] \to \mathbb{R}$ are continuous with $e^{i\psi(t)} = e^{i\phi(t)}$ for all t. Then $\frac{\psi - \phi}{2\pi}$ is a continuous function on $[0,1]$ taking values in $\mathbb{Z}$; since $[0,1]$ is connected and $\mathbb{Z}$ is discrete, it is constant. If in addition $\psi(0) = \phi(0)$, the constant is 0. □

Corollary 14.2

If $\gamma : [0,1] \to \mathbb{C} \setminus \{0\}$ is continuous with $\gamma(0) = |\gamma(0)|e^{i\theta_0}$, then there is a unique continuous $\theta : [0,1] \to \mathbb{R}$ with $\theta(0) = \theta_0$ and $\gamma(t) = |\gamma(t)|e^{i\theta(t)}$.

Proof. Apply Theorem 14.1 to $g(t) = \gamma(t)/|\gamma(t)|$, which is continuous since γ never vanishes. □

Definition 14.3 (Winding Number)

Let $\gamma : [0,1] \to \mathbb{C} \setminus \{0\}$ be continuous and let θ be as in Corollary 14.2. We define the **winding number** of γ about 0 to be

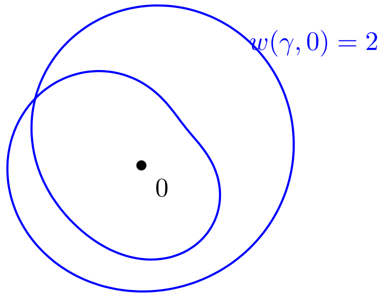
$$w(\gamma, 0) = \frac{\theta(1) - \theta(0)}{2\pi}.$$

By the uniqueness in Theorem 14.1, any two choices of θ differ by a constant, so $w(\gamma, 0)$ is well defined. From now on we shall only be interested in **closed** curves, that is those with $\gamma(0) = \gamma(1)$.

Lemma 14.4

If γ is closed then $w(\gamma, 0) \in \mathbb{Z}$, and every integer arises.

Proof. $\gamma(0) = \gamma(1)$ gives $e^{i\theta(0)} = e^{i\theta(1)}$, so $\theta(1) - \theta(0) \in 2\pi\mathbb{Z}$. For the converse take $\gamma(t) = e^{2\pi i n t}$, for which $\theta(t) = 2\pi n t$ and $w(\gamma, 0) = n$. □



Winding numbers behave exactly as one would hope under multiplication.

Lemma 14.5

If $\gamma_1, \gamma_2 : [0,1] \to \mathbb{C} \setminus \{0\}$ are continuous, then $w(\gamma_1\gamma_2, 0) = w(\gamma_1, 0) + w(\gamma_2, 0)$.

Proof. If $\gamma_j(t) = |\gamma_j(t)|e^{i\theta_j(t)}$ with θ_j continuous, then $\gamma_1(t)\gamma_2(t) = |\gamma_1(t)\gamma_2(t)|e^{i(\theta_1(t)+\theta_2(t))}$ and $\theta_1 + \theta_2$ is continuous, so it is a valid choice of angle function for $\gamma_1\gamma_2$. $\square$

§14.2 Walking the Dog

Lemma 14.6 (Dog Walking Lemma)

Let $\gamma_1, \gamma_2 : [0,1] \to \mathbb{C}$ be continuous closed curves with $\gamma_1(t) \neq 0$ and $|\gamma_2(t)| < |\gamma_1(t)|$ for all t. Then $\gamma_1 + \gamma_2$ never vanishes and

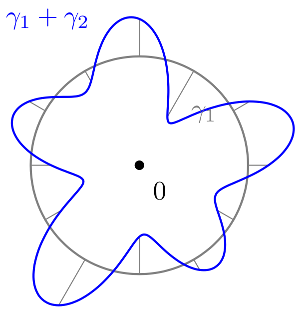
$$w(\gamma_1 + \gamma_2, 0) = w(\gamma_1, 0).$$

Proof. Write $\gamma_1 + \gamma_2 = \gamma_1\left(1 + \frac{\gamma_2}{\gamma_1}\right)$. Since $\left|\frac{\gamma_2(t)}{\gamma_1(t)}\right| < 1$, the curve $\eta = 1 + \frac{\gamma_2}{\gamma_1}$ takes values in $\{z : \operatorname{Re} z > 0\}$, and in particular never vanishes; hence neither does $\gamma_1 + \gamma_2$. By Lemma 14.5,

$$w(\gamma_1 + \gamma_2, 0) = w(\gamma_1, 0) + w(\eta, 0),$$

so it suffices to show $w(\eta, 0) = 0$. But on $\{\operatorname{Re} z > 0\}$ we may take $\theta(t) = \arg \eta(t) \in (-\frac{\pi}{2}, \frac{\pi}{2})$ as our continuous angle function, so $|\theta(1) - \theta(0)| < \pi$; as η is closed, $\theta(1) - \theta(0) \in 2\pi\mathbb{Z}$, forcing $\theta(1) = \theta(0)$. $\square$

The name comes from the following picture. A man walks round a closed path γ_1 which avoids a lamp-post at the origin, taking his dog with him on a lead of length always shorter than his own distance from the lamp-post. The dog's path is $\gamma_1 + \gamma_2$. Then, however much the dog dashes about, man and dog go round the lamp-post the same number of times.



The lemma is really a statement about small perturbations. The right way to say 'small perturbation' in general is the following.

Definition 14.7 (Homotopy)
Let $\gamma_0, \gamma_1 : [0,1] \to \mathbb{C} \setminus \{0\}$ be closed curves. We say γ_0 and γ_1 are **homotopic** through closed curves avoiding 0 if there is a continuous $\Gamma : [0,1]^2 \to \mathbb{C} \setminus \{0\}$ with

$$\Gamma(s,0) = \Gamma(s,1) \ \ \forall s, \qquad \Gamma(0,t) = \gamma_0(t) \ \ \forall t, \qquad \Gamma(1,t) = \gamma_1(t) \ \ \forall t.$$

We write $\gamma_0 \simeq \gamma_1$; it is easy to check that $\simeq$ is an equivalence relation.

Theorem 14.8 (Dog Walking Along a Canal)
If $\gamma_0 \simeq \gamma_1$ then $w(\gamma_0, 0) = w(\gamma_1, 0)$.

Proof. Let Γ be a homotopy. Since $|\Gamma|$ is continuous and strictly positive on the compact set $[0,1]^2$, it attains a strictly positive minimum, so there is a $\delta > 0$ with $|\Gamma(s,t)| \geq 8\delta$ everywhere. Also Γ is uniformly continuous, so we may find N with

$$|\Gamma(s,t) - \Gamma(s',t')| < \delta \qquad \text{whenever } |s-s'|, |t-t'| \leq \tfrac{1}{N}.$$

Set $\tau_r(t) = \Gamma(r/N, t)$ for $0 \leq r \leq N$, a closed curve avoiding 0. Then for every t,

$$|\tau_{r+1}(t) - \tau_r(t)| < \delta < 8\delta \leq |\tau_r(t)|,$$

so by the dog walking Lemma 14.6 (with $\gamma_1 = \tau_r$ and $\gamma_2 = \tau_{r+1} - \tau_r$) we get $w(\tau_{r+1}, 0) = w(\tau_r, 0)$. Chaining these,

$$w(\gamma_0, 0) = w(\tau_0, 0) = w(\tau_1, 0) = \cdots = w(\tau_N, 0) = w(\gamma_1, 0). \qquad \square$$

§14.3 Two Old Theorems Again

The following is the workhorse: a non-zero winding number on the boundary forces a zero inside.

Corollary 14.9
Let $f : \overline{B}(0,R) \to \mathbb{C}$ be continuous with $f(z) \neq 0$ for $|z| = R$, and define $\gamma(t) = f(Re^{2\pi i t})$. If $w(\gamma, 0) \neq 0$ then $f(z) = 0$ for some z with $|z| < R$.

Proof. Suppose f has no zero in $\overline{B}(0,R)$. Then $\Gamma(s,t) = f(sRe^{2\pi i t})$ is a continuous map $[0,1]^2 \to \mathbb{C} \setminus \{0\}$ with $\Gamma(s,0) = \Gamma(s,1)$, so it is a homotopy from the constant curve $t \mapsto f(0)$ to γ. The constant curve has winding number 0, so by Theorem 14.8 $w(\gamma, 0) = 0$, a contradiction. $\square$

Theorem 14.10 (Fundamental Theorem of Algebra, again)
Every non-constant complex polynomial has a root.

Proof. We may assume $P(z) = z^n + \sum_{j=0}^{n-1} a_j z^j$ with $n \geq 1$. Choose $R > \max\{1, \sum_{j=0}^{n-1} |a_j|\}$ and set $\gamma_1(t) = R^n e^{2\pi int}$, $\gamma_2(t) = \sum_{j=0}^{n-1} a_j R^j e^{2\pi ijt}$, so that $\gamma_1 + \gamma_2$ is the curve $t \mapsto P(Re^{2\pi it})$. For every t,

$$|\gamma_2(t)| \leq \sum_{j=0}^{n-1} |a_j| R^j \leq R^{n-1} \sum_{j=0}^{n-1} |a_j| < R^n = |\gamma_1(t)|,$$

so by Lemma 14.6 we get $w(\gamma_1 + \gamma_2, 0) = w(\gamma_1, 0) = n \neq 0$. Now Corollary 14.9 gives a root in $B(0, R)$. □

This is the same idea as our very first proof in Theorem 1.14, but the bookkeeping is now done by the winding number rather than by hand. Finally, we recover the no retraction theorem, and with it Brouwer's fixed point theorem in two dimensions.

Theorem 14.11

There is no continuous $f : \overline{D} \to \partial D$ with $f(z) = z$ for all $z \in \partial D$.

Proof. Suppose such an f existed and set $\Gamma(s,t) = f(se^{2\pi it})$ for $s, t \in [0,1]$. This is continuous, satisfies $\Gamma(s,0) = \Gamma(s,1)$, and never vanishes since $|f| \equiv 1$ on ∂D. So Γ is a homotopy through closed curves avoiding 0 between $t \mapsto \Gamma(0,t) = f(0)$, a constant curve of winding number 0, and $t \mapsto \Gamma(1,t) = f(e^{2\pi it}) = e^{2\pi it}$, which has winding number 1. This contradicts Theorem 14.8. □

It is worth comparing the two proofs of the no retraction theorem. The one via Sperner's lemma is combinatorial, elementary, and generalises to $\mathbb{R}^n$ with only cosmetic changes; the one just given is slicker, but is tied to the plane, because the whole notion of a winding number is.

More generally, the contents of this section show that some of what one learns in a complex analysis course – the argument principle, Rouché's theorem, the fundamental theorem of algebra – is really a special case of general topological facts about continuous maps, and has nothing to do with differentiability. Other parts of the subject, such as Taylor's theorem and Cauchy's theorem itself, depend crucially on the fact that we are dealing with the very restricted class of functions satisfying the Cauchy–Riemann equations. In traditional courses this distinction appears late, if at all; Beardon's *Complex Analysis* shows that it is possible to do things differently, and is well worth a look.